



PROVISIONING AND INSTALLATION GUIDE OF THE K2VIEW AZURE MARKETPLACE APPLICATION

K2view Data Product Platform Services

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Overview

This guide provides a comprehensive walkthrough for provisioning and installing the K2view Data Product Platform in a self-hosted environment on Microsoft Azure. Targeted for IT and DevOps audiences, it details each phase of the installation process - from the steps and prerequisites to install the K2view Azure Application resources to deploying and configuring the K2view's cloud services within your self-hosted environment. The guide addresses setting up Kubernetes clusters, managing resource groups, and ensuring connectivity to customer-managed services.

This documentation outlines necessary prerequisites, roles, permissions, and post-installation validation steps. Please use the prerequisites and installation checklist to prepare and plan your installation.

Should a problem arise during the installation, please consult with your K2view representative to obtain support.

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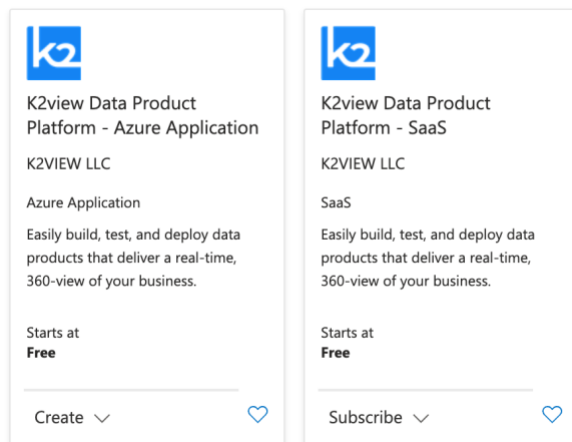
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1. The End State

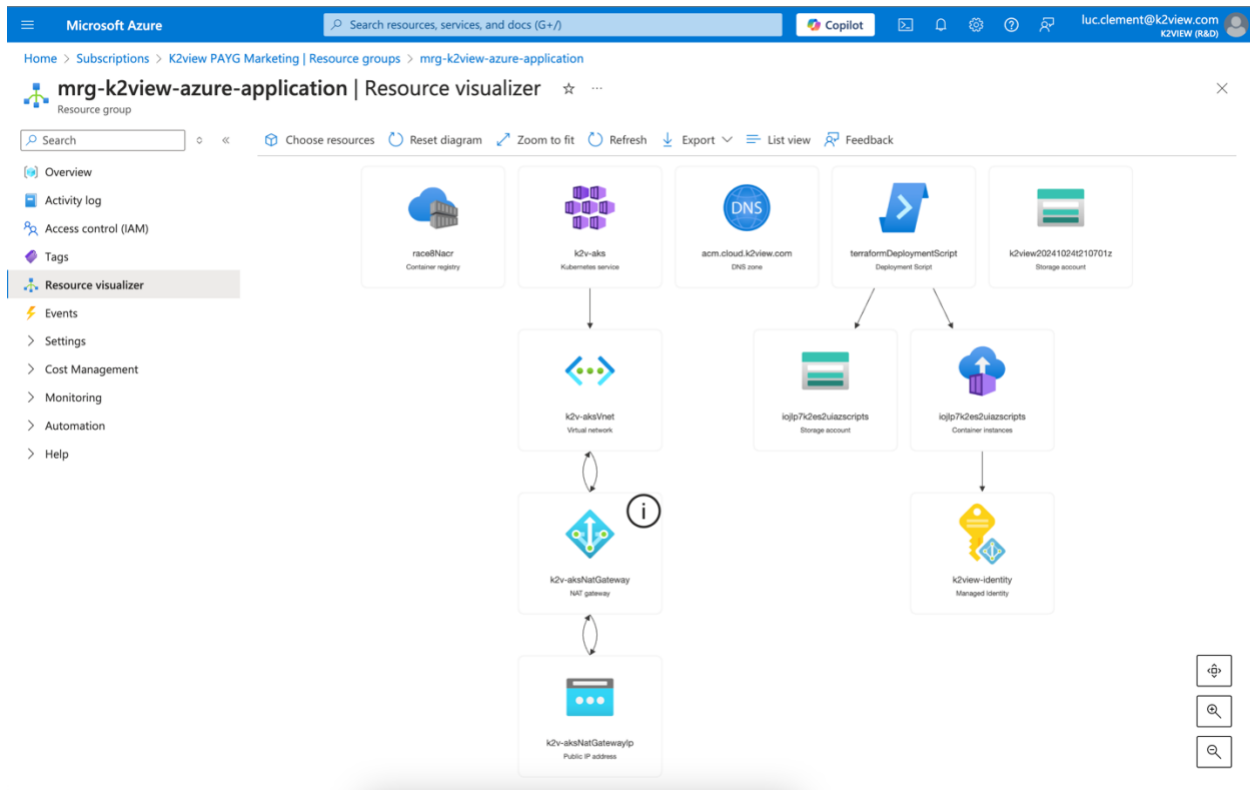
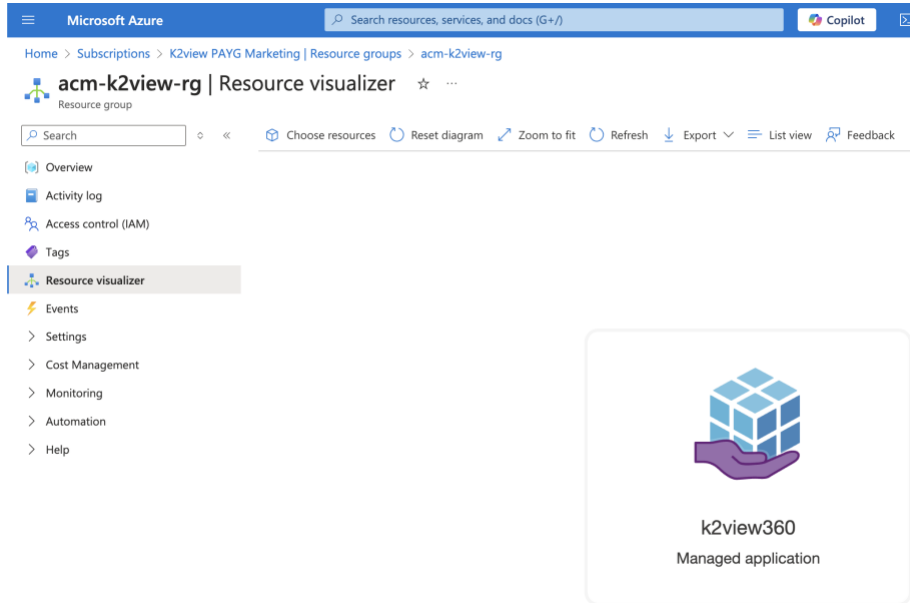
Upon completing the instructions in this guide, the K2view Data Product Platform will be fully deployed and operational within your Azure environment. This includes a fully configured Azure Kubernetes Service (AKS) cluster, secured network components, and connected resource groups, allowing you to start building and managing data products. The installed environment will be ready for integration with your existing data sources and K2view's cloud services, enabling the creation of projects and spaces for data management. Additionally, your Azure Container Registry will be populated with the necessary K2view Docker images, setting the foundation for scalable and secure operations

The [Planning](#) section will guide you with the information you need to gather to complete the installation. Having everything you need on hand will help ensure a fast and smooth installation and configuration.

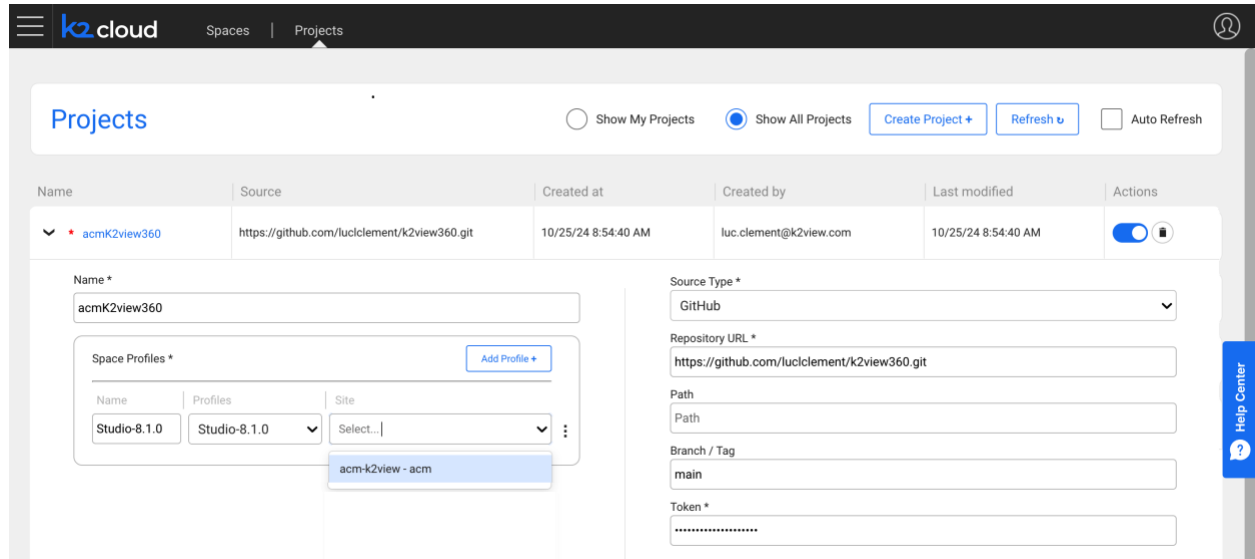
The process start by selecting the **K2view Data Product Platform—Azure Application** listing on the Microsoft Azure Marketplace. This listing will be used to install the K2view Data Product Platform within your Azure environment.



Upon completing the installation two resource groups will have been created: one for the K2view Managed Application and the other for the Kubernetes and infrastructure services on which the K2view Data Product Platform services will run.



Upon completion of the installation, you will create your K2view project.



The screenshot shows the 'Projects' page in the K2view cloud interface. The project 'acmk2view360' is selected. The details form shows the following information:

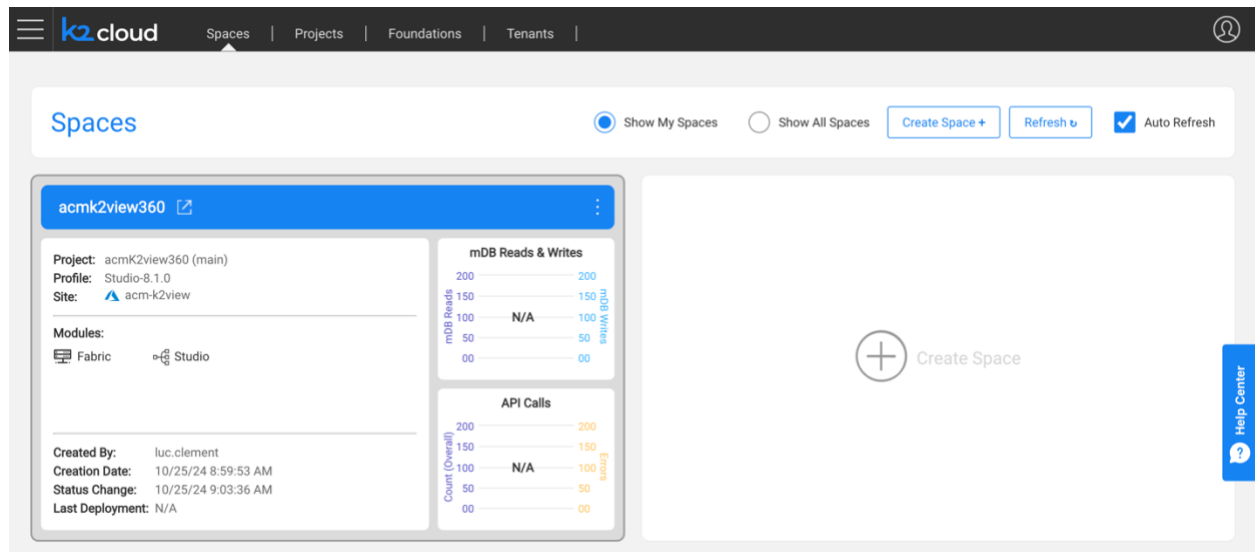
- Name:** acmk2view360
- Source:** https://github.com/luccllement/k2view360.git
- Created at:** 10/25/24 8:54:40 AM
- Created by:** luc.clement@k2view.com
- Last modified:** 10/25/24 8:54:40 AM
- Source Type:** GitHub
- Repository URL:** https://github.com/luccllement/k2view360.git
- Path:** Path
- Branch / Tag:** main
- Token:** *****

Below the project details, there is a 'Space Profiles' section with a table showing the selected profile 'Studio-8.1.0' and site 'acm-k2view - acm'.

Name	Profiles	Site
Studio-8.1.0	Studio-8.1.0	Select...

The 'acm-k2view - acm' profile is highlighted in the dropdown menu.

And then, you will create your first K2view Space.



The screenshot shows the 'Spaces' page in the K2view cloud interface. The space 'acmk2view360' is selected. The details form shows the following information:

- Project:** acmk2view360 (main)
- Profile:** Studio-8.1.0
- Site:** acm-k2view
- Modules:** Fabric, Studio
- Created By:** luc.clement
- Creation Date:** 10/25/24 8:59:53 AM
- Status Change:** 10/25/24 9:03:36 AM
- Last Deployment:** N/A

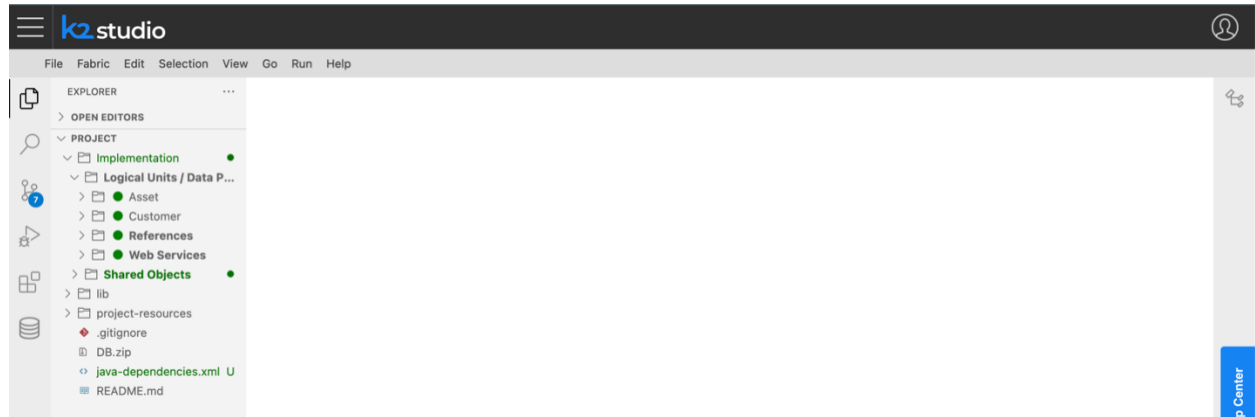
Below the project details, there are two charts: 'mDB Reads & Writes' and 'API Calls'. Both charts show 'N/A' for the current data.

The 'mDB Reads & Writes' chart shows a scale from 0 to 200 for both reads and writes.

The 'API Calls' chart shows a scale from 0 to 200 for the count (overall) and errors.

On the right side of the space details, there is a large 'Create Space' button with a plus icon.

And finally, you will be ready to start your first project, and start learning using our Knowledge Base at <https://support.k2view.com/knowledge-base.html>, where you can find documentation and demo projects, and sign up for the K2view Academy Learning Center at <https://academy.k2view.com>.



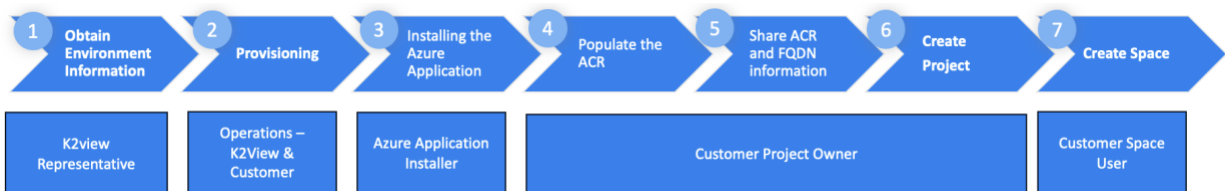
Next, let's plan this installation.

2. Planning

Planning and capturing the necessary information prior to the installation will help ensure a smooth installation experience. The customer and K2view will gather and share information used to provision the environment.

This section summarizes the steps, constraints, and prerequisites. At the end of the section, you will find an [Installation Checklist](#) that will help you ensure you have everything on hand to complete the installation.

2.1. Steps



Step 1 – Obtaining Environment Information

A K2view representative working with the customer will discuss the prerequisites to complete the installation and review the [Installation Checklist](#).

Step 2 - Provisioning

During the provisioning step:

- K2view will create Nexus and user accounts, create a tenant, and create associated configurations. Some of the information that K2view needs to complete the configuration will need to be provided by the customer in Step 5.
- The customer will gather information based found in the [Installation Checklist](#) such as Git details, a TLS certificate (PEM encoded) and its private key (PEM encoded), and names of entities. The certificate and private key files need to be base64 encoded. The customer will also need to validate prerequisites and created permissions for the user installing the K2view Azure Application within the Azure subscription.

Step 3 – Install the K2view Azure Application

A person with the necessary roles and permissions within the Azure subscription will run the [K2view Azure Application installation](#).

Step 4 – Populate the Azure Container Registry with K2view Docker images

The K2view project owner will [load the selected Azure Container Registry](#) with K2view Docker images that have been identified as required for the customer during Step 1.

Step 5 – Share ACR and Domain Name information with K2view

To complete the configuration for the tenant's creation of Projects and Spaces, the Customer Project Owner will need to share the following information used.

- The fully qualified domain name (FQDN) used for your Domain Ingress
- The list of Azure Container Registry (ACR) locations of your Docker images
- The AKS Cluster name

This information may have been identified during pre-planning, but the specific information should be confirmed to avoid misconfiguration.

Step 6 - [Create the K2view Project](#)

Step 7 - [Create a K2view Space](#)

2.2. Constraints

There are three constraints you need to be aware of to install the K2view Azure Application within your Azure subscription.

- **Azure Subscription Type.** You will be installing the K2view Azure Application in an Azure Subscription that needs to be a Pay-As-You-Go (PAYG) or Enterprise Agreement subscription.
- **Subscription Role and Permission.** The installer of the K2view Azure Application needs specific Subscription level roles and permissions.
 - The user needs to be a Subscription Owner.
 - The user performing the K2view Azure Application installation requires the Microsoft.Authorization roleAssignments of write, read, and delete within the subscription. We describe how to [create a custom role](#) for this purpose.
- **Region and availability zones.** Please note that the installation requires you select an Azure region that offers more than one availability zones. Not all Azure Regions support multiple availability zone.

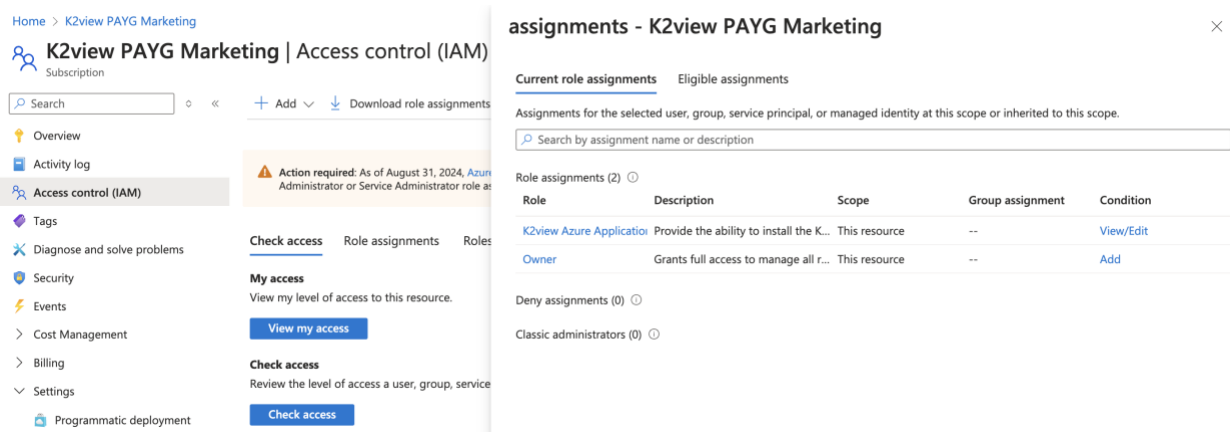
2.3. Prerequisites

Certain prerequisites must be met, and planning steps must be taken to install the K2view Azure Application, set up the Azure Container Registry, and create a K2view Project and a Space.

These prerequisites are summarized here.

- You need to verify that your Azure Subscription is a PAYG or an Enterprise Agreement subscription.
- You need the person performing the K2view Azure Application installation to have a Subscription-level Owner role and a custom role that grants specific permissions. This user needs the Subscription Owner Role with a [Custom Role](#) that grants the user
 1. "Microsoft.Authorization/roleAssignments/write",
 2. "Microsoft.Authorization/roleAssignments/read",
 3. "Microsoft.Authorization/roleAssignments/delete"

The role assignment once created is depicted here. The [Permissions and Custom Role Creation](#) section describes how to create a custom role with the associated permissions.



The screenshot displays the Azure portal interface for 'K2view PAYG Marketing | Access control (IAM)'. The left-hand navigation pane includes links for Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Security, Events, Cost Management, Billing, Settings, and Programmatic deployment. The 'Access control (IAM)' section is active, showing a search bar, '+ Add' button, and 'Download role assignments' link. Below this, there are sections for 'Check access', 'My access', and 'Check access' with corresponding buttons. The main content area is titled 'assignments - K2view PAYG Marketing' and shows 'Current role assignments' and 'Eligible assignments'. A table lists the current role assignments:

Role	Description	Scope	Group assignment	Condition
K2view Azure Application	Provide the ability to install the K...	This resource	--	View/Edit
Owner	Grants full access to manage all r...	This resource	--	Add

Below the table, there are sections for 'Deny assignments (0)' and 'Classic administrators (0)'.

- A Git code repository is required for your Fabric Project. Three pieces of information will be required to create your K2view project including:
 - Repository URL
 - Git token
 - Git Branch / Tag

The creation of a Git repository, if one is not already available for your use, is described in the [GitHub](#) section.

- You will need access to K2view's Nexus Repository and Docker installed on a machine used to populate the Azure Container Registry (ACR) that will be created within your Azure Resource Group.
 - The installation of the K2view Azure Application will create an ACR; if you plan to use a pre-existing ACR you may do so. Whether you use the ACR or an OCI-compatible

container registry the location where you will be populating K2view Docker images needs to be provided to K2view as described in Step 5 above and in the [Sharing information with K2view](#) section.

- You need to populate your selected ACR by pulling and pushing docker images.
 1. Docker needs to be installed on a computer to pull and push docker images from the K2view Nexus repository to the ACR created
 2. You will need a K2view Nexus account you can obtain from your K2view representative

The [Populating Your Azure Container Registry](#) section describes the steps.

- Pre-planning names will help a smooth installation. Use the [Checklist](#) and capture this information in a document. This includes:
 - An Azure region to be used. Please note that the installation requires an Azure Region that provides two or more availability zones. Not all Azure Regions have multiple availability zones.
 - The names of the two resource groups where K2view Fabric services will be installed.
 1. **Azure Application Resource Group.** You can choose an existing resource group or create a new one. You should select a representative name. In the example demonstrated in this guide we use “acm-k2view-rg”. Select something that has significance to your project.
 2. **Azure Managed Application Resource Group.** K2view resources will be created in a separate resource group, the Azure Managed Application resource group. This resource group contains K2view infrastructure components created during the installation. A name will be suggested for you. You can use that name or one that has significance to you. In our example, we used **mrg-k2view-azure-application**.
 - **Cluster Name:** This is an Azure Kubernetes Service (AKS) cluster name of the Kubernetes cluster to be created. The default, if used, is k2v-aks.
 - **Environment Tag:** This tag describes the project's environment. Typical values could be Dev, UAT, and Prod. Azure cost management modules will use this tag to help you categorize costs.
 - **Project Tag:** This tag describes the functional purpose of the project. You might consider tag names such as K2view TDM, K2view Masking, and K2view 360. Azure cost management modules will use this tag to help you categorize costs.
 - **Domain Ingress:** This is the FQDN of the Nginx Ingress Controller for which a PEM wildcard certificate needs to be created. In our example, we used acm.cloud.k2view.com, a domain owned by K2view. You will need to select your own domain and one for which you can create TLS certificates (PEM encoded).

- TLS keys and certificates

Your network team must create a wildcard TLS certificate for the domain you selected as your Domain Ingress. The full certificate authority chain needs to be present. During the installation, you will need to paste the base64 values of your PEM TLS certificate and its associated private key.

- **TLS Private Key:** the base64-encoded PEM private key associated with the TLS certificate
- **TLS Certificate:** a base64-encoded PEM wildcard certificate. For example, for the domain ingress of acm.cloud.k2view.com you need to create a wildcard certificate for *.acm.cloud.k2view.com for hosts of this domain.

These should look like the following, respectively (shortened for brevity and non-disclosure of sensitive):

PEM TLS Private Key:

```
-----BEGIN PRIVATE KEY-----
MIIJQwIBADANBgkqhkiG9w0BAQEFAASCCS0wgGkpAgEAAoICAQC1CFTpFyQaTg1b
...
fHZLomEYrUhsV9X+RUz6f3WnzzbfXCA=
-----END PRIVATE KEY-----
```

PEM TLS Certificate:

```
-----BEGIN CERTIFICATE-----
MIIFUjCCAzOCCQDjx3pKVc6ytTANBgkqhkiG9w0BAQsFADBhMQswCQYDVQQGEwJV
...
YfROGHT81E1aDUh17MHvRbnHI4qekw==
-----END CERTIFICATE-----
```

The PEM certificate and key must be further encoded using base64 as input to the installation screen. To encode a PEM certificate (e.g., cert.pem) to base64 on both Linux and Windows, you can use the following commands.

Linux: You can use the base64 command to read the cert.pem file, encodes it in base64, and saves the output to cert_base64.txt.

```
base64 cert.pem > cert_base64.txt
```

Windows: On Windows, you can use the certutil command to encode the cert.pem file in base64 and saves the result to cert_base64.txt.

```
certutil -encode cert.pem cert_base64.txt
```

The base64-encoded versions of the private key and certificate should look like the following:

base64-encoded TLS Private Key:

```
LS0tLS1CRUdJTiBQUk1WQVRFIEtFW50tLS0tCk1JSUprRd01CQURBTk1na3Foa21HOXcwQkFRRUZBQVNDQ1  
Mwd2dna3BBZ0VBQW9JQ0FRRE9yakUrNONQVku2QVgKV01FRFVvdnVlc01XOEJ4MTR  
L...  
QT1ZV1pKKz1qRkhFUTdBZXB3S1RtMEVraz0KLS0tLS1FTkQgUUFJJVkfURSBURVktLS0tLQo=
```

base64-encoded TLS Certificate:

```
LS0tLS1CRUdJTiBDRVJUSUZJQ0FURS0tLS0tCk1JSUdGRENDQ1B5Z0F3SUJBZ01TQS9YVFJKAUpXOXMzb1  
dZZlN5aORpZ1h5TUEwR0NTcUdTSWZzRFFFQkN3VUEKTURNeEN6QUUpCZ05WQkFZVEFsV1RNU113RkFZRFZR  
UUtFdzFNW1hRbm  
...  
1ND0KLS0tLS1FTkQgQ0VSVE1GSUNBVEU1LS0tLQ==
```

Prerequisite: You should prepare these base64 encoded files before the start of the installation.

- You need to have been provisioned by K2view. Provisioning includes:
 - Creation of a tenant and configurations for you
 - Creation of an initial K2view Admin user
 - Creation of a K2view Nexus repository account
 - Creation of a K2view Cloud Mailbox ID

In consultation with K2view, Docker images and their associated versions will be recommended. You will need this information to use docker pull commands to obtain images such as fabric, fabric-studio, and the k2-cloud-deployer images. These commands are described in the [Populating Your Azure Container Registry](#) section.

K2view may provide you with access to k2exchange.k2view.com optionally. Please discuss your needs with your K2view representative.

- After installing your K2view Azure Application, you will need to pause to provide K2view with information that will allow you to finalize your environment.
 - After you complete the installation and before you create a K2view project and a space, you will need to provide K2view with the location images within your container registry.
 - The list of images is those you will push as described in the [Commands to Push Containers to the Azure Container Registry](#)
 - For example, if you pushed fabric-studio:8.1.0_199 to the race8nacr.azurecr.io container registry used in this guide's examples, you will need to provide K2view with this value: race8nacr.azurecr.io/k2view/fabric-studio:8.1.0_199
 - Provide the location for every image pushed to your container registry.

2.4. Installation Checklist

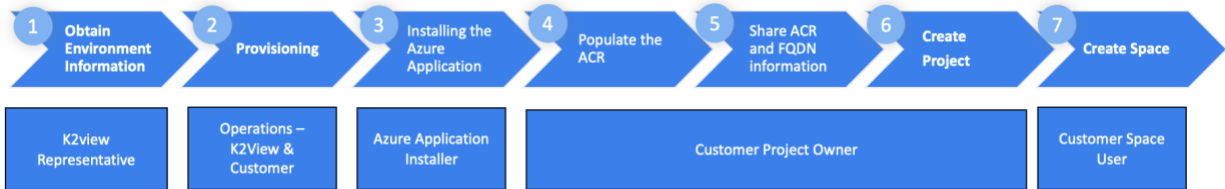
Here is a checklist to help you gather the values and content you should have before the installation:

1. Identify an individual who will install the K2view Azure Application. This individual needs to hold a Subscription Owner role and specific permissions via the creation of an Azure custom role – see [Permissions and Custom Role Creation](#).
2. Identify the Azure Region you will be installing the K2view Azure Application, and confirm that it supports multiple availability zones
3. During the installation, you will need to enter the following:
 - a. An Azure Application Resource Group name
 - b. An Azure Managed Application Resource Group name
 - c. An AKS Cluster name
 - d. An Environment Tag name
 - e. A Project Tag name
 - f. Your Domain Ingress FQDN
 - g. Obtained a Wildcard TLS PEM certificate for the Domain Ingress FQDN
 - h. Obtained the corresponding PEM private key
 - i. Base64-encode both the certificate and private key files
 - j. The K2view Cloud Mailbox ID provided to you by K2view
4. To populate your Azure Container Repository, you will need
 - a. K2view Nexus Repository Account supplied to you by K2view
 - b. Obtained the list of Docker images to pull from Nexus and push to in consultation with K2view
 - c. Docker installed on the computer used to pull and push Docker images to your ACR
5. After the image population of your ACR is complete, and before the creation of your K2view Project and your first Space, you need to provide K2view with. Once confirmed you will be able to create your K2view Project:
 - a. The FQDN of your Domain Ingress
 - b. The list of ACR locations of your Docker images
 - c. The AKS Cluster name
6. To create your K2view project, you will need:
 - a. A Git Repository URL
 - b. Git token
 - c. Git Branch / Tag
 - d. Your initial K2view Admin user

2.5. Post Installation

- Validation of the environment should be performed, including the DNS zone configuration. Please see the [DNS zone](#) section.
- Should SSO and IDP federation be desired, following the initial installation, additional information will be required and planned in collaboration with K2view.

3. Installing the K2view Azure Application



This section describes Step 3 – The Installation of the K2view Azure Application.

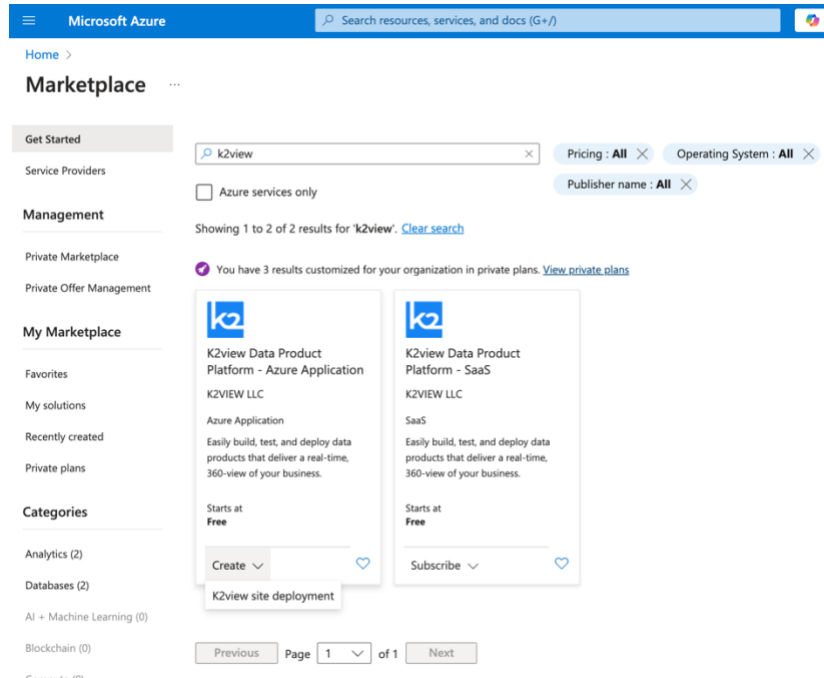
The prerequisites of this section are:

1. Identify an individual who will install the K2view Azure Application. This individual needs to hold a Subscription Owner role and specific permissions via the creation of an Azure custom role—see [Permissions and Custom Role Creation](#).
2. Identify the Azure Region you will be installing the K2view Azure Application, and confirm that it supports multiple availability zones.
3. During the installation, you will need to enter the following:
 - a. An Azure Application Resource Group name
 - b. An Azure Managed Application Resource Group name
 - c. An AKS Cluster name
 - d. An Environment Tag name
 - e. A Project Tag name
 - f. Your Domain Ingress FQDN
 - g. Obtained a Wildcard TLS PEM certificate for the Domain Ingress FQDN
 - h. Obtained the corresponding PEM private key
 - i. Base64 encoded both versions of the certificate and private key files
 - j. The K2view Cloud Mailbox ID provided to you by K2view

The Azure Application installer is available and found on the Azure Marketplace. Search for “K2view” in the marketplace as shown below. For the purpose of this installation you will use the “K2view Data Product Platform – Azure Application.”

This application will fully install Fabric services within a resource group of your Azure subscription. The installation process is fully automated and requires only minimal input provided in the [Installation Checklist](#) section. Please review the [Planning](#) in its entirety to understand the constraints, prerequisites, and the meaning of the terms used.

Search for K2view

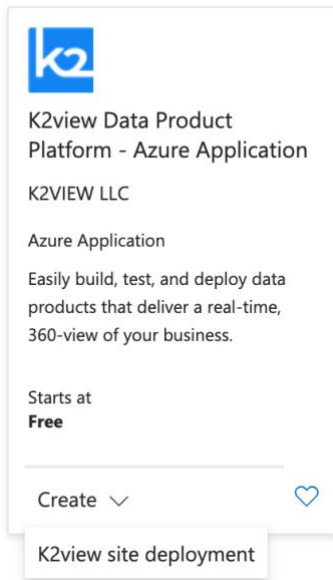


The screenshot shows the Microsoft Azure Marketplace interface. At the top, there's a search bar with 'k2view' entered. Below the search bar, there are filters for 'Pricing: All', 'Operating System: All', and 'Publisher name: All'. A checkbox for 'Azure services only' is also present. The search results show two items:

- K2view Data Product Platform - Azure Application** by K2VIEW LLC. It is an Azure Application that allows users to easily build, test, and deploy data products that deliver a real-time, 360-view of their business. It starts at Free. A 'Create' button is visible, with a tooltip that says 'K2view site deployment'.
- K2view Data Product Platform - SaaS** by K2VIEW LLC. It is a SaaS application that allows users to easily build, test, and deploy data products that deliver a real-time, 360-view of their business. It starts at Free. A 'Subscribe' button is visible.

At the bottom, there are pagination controls showing 'Page 1 of 1'.

Select the Azure Application listing and press Create.



The installation process will guide you through several screens. You must select the appropriate Azure subscription, a resource group that can be a preexisting one or one you create, and the region where the Fabric services will be installed. Additionally, you must provide the name of the Azure-managed application and its associated resource group.

Please consult the [Constraints](#) and [Prerequisites](#) section to ensure you have the permissions and identified an Azure region that supports multiple availability zones.

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

luc.clement@k2view.com
K2VIEW (R&D)

[Home](#) > [Marketplace](#) >

Create K2view site deployment

Basics

K2view Installation settings

K2view Application settings

Review + create

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ

K2view PAYG Marketing

Resource group * ⓘ

Create new

Instance details

Region ⓘ

Loading...

Managed Application Details

Provide a name for your managed application, and its managed resource group. Your application's managed resource group holds all the resources that are required by the managed application.

Application Name *

Managed Resource Group * ⓘ

mrg-k2view-azure-application-20241024140403

Previous

Next

Review + create

Give feedback

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

luc.clement@k2view.com
K2VIEW (R&D)

[Home](#) > [Marketplace](#) >

Create K2view site deployment

Basics

K2view Installation settings

K2view Application settings

Review + create

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ

K2view PAYG Marketing

Resource group * ⓘ

acm-k2view-rg

Create new

Instance details

Region * ⓘ

East US 2

Managed Application Details

Provide a name for your managed application, and its managed resource group. Your application's managed resource group holds all the resources that are required by the managed application.

Application Name *

k2view360

Managed Resource Group * ⓘ

mrg-k2view-azure-application

Previous

Next

Review + create

Give feedback

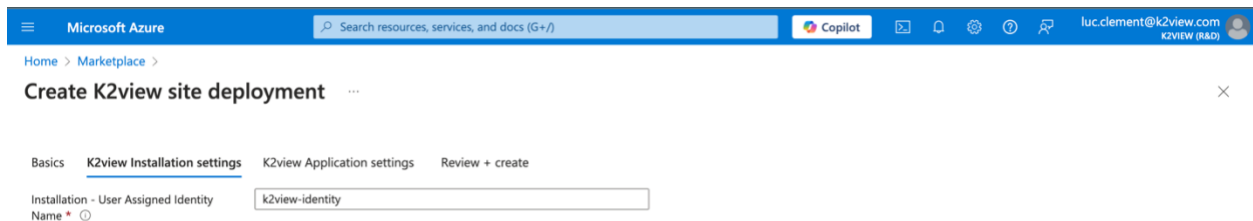
The installation uses two resource groups.

- **Azure Application Resource Group.** This is shown on the page below “Subscription”. You can choose an existing resource group or create a new one. You should select a representative name. In the example demonstrated in this guide we use “acm-k2view-rg”. Select something that has significance to your project.
- **Azure Managed Application Resource Group.** This is shown on the page as “Managed Resource Group”. K2view resources will be created in a separate resource group, the Azure Managed Application resource group. This resource group contains K2view infrastructure components created during the installation. A name will be suggested for you. You can use that name or one that has significance to you. In our example, we used **mrg-k2view-azure-application**.

Press Next at the bottom of the page.

Second Page of the Installation Wizard

On this screen you set the identity used for the installation. Unless you have specific needs to do otherwise, we recommend that you use the default of “k2view-identity”

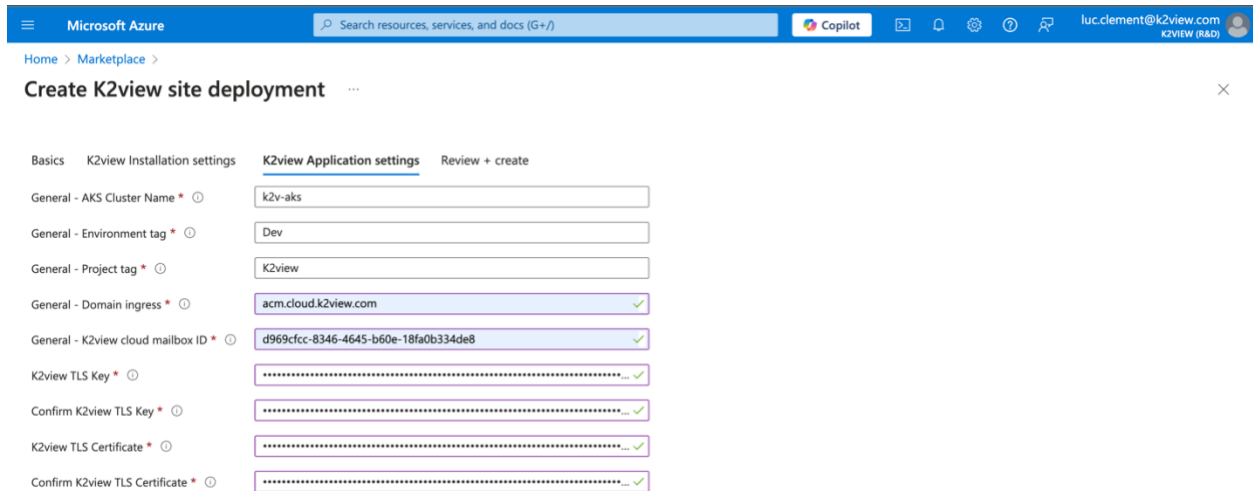


The screenshot shows the 'Create K2view site deployment' wizard in the Microsoft Azure portal. The 'K2view Installation settings' tab is active. Under 'Installation - User Assigned Identity', the 'Name' field is set to 'k2view-identity'. The breadcrumb navigation shows 'Home > Marketplace > Create K2view site deployment'. The top navigation bar includes the Microsoft Azure logo, a search bar, Copilot, and the user profile 'luc.clement@k2view.com'.

Press Next at the bottom of the page.

Third Page of the Installation Wizard

The third step of the process is where you provide the details you will find in the [Installation Checklist](#).



Microsoft Azure | Search resources, services, and docs (G+/I) | Copilot | luc.clement@k2view.com | K2VIEW (RAO)

Home > Marketplace > Create K2view site deployment

Basics | K2view Installation settings | **K2view Application settings** | Review + create

General - AKS Cluster Name *

General - Environment tag *

General - Project tag *

General - Domain ingress *

General - K2view cloud mailbox ID *

K2view TLS Key *

Confirm K2view TLS Key *

K2view TLS Certificate *

Confirm K2view TLS Certificate *

Please enter

- **Cluster Name:** This is an Azure Kubernetes Service (AKS) cluster name of the Kubernetes cluster to be created. The default, if used, is k2v-aks.
- **Environment Tag:** This tag describes the project's environment. Typical values could be Dev, UAT, and Prod. Azure cost management modules will use this tag to help you categorize costs.
- **Project Tag:** This tag describes the functional purpose of the project. You might consider tag names such as K2view TDM, K2view Masking, and K2view 360. Azure cost management modules will use this tag to help you categorize costs.
- **Domain Ingress:** This is the FQDN of the Nginx Ingress Controller for which a wildcard certificate needs to be created. In our example, we used acm.cloud.k2view.com, a domain owned by K2view. You will need to select your own domain and one for which you can create TLS certificates (PEM encoded).
- **TK2view Cloud Mailbox ID** provided to you by K2view
- **The base64-encoded values of your PEM TLS certificate and its associated private key**

Your network team must create a wildcard TLS certificate for the domain you selected as your Domain Ingress. The full certificate authority chain needs to be present. During the installation, you will need to paste the base64-encoded values of your TLS certificate and its associated private key.

- **TLS Private:** the base64-encoded private key associated with the TLS certificate
- **TLS Certificate:** a base64-encoded wildcard certificate. For example, for the domain ingress of acm.cloud.k2view.com you need to create a wildcard certificate for *.acm.cloud.k2view.com for hosts of this domain.

What you paste in should look like the following:

base64-encoded TLS Private Key:

```
LS0tLS1CRUdJTiBQUk1wQVRFIEtFWS0tLS0tCk1JSUpRd01CQURBTk1na3Foa21HOXcwQkFRRUZBQVNDQ1  
Mwd2dna3BBZ0VBQW9JQ0FRRE9yakUrNONQVku2QVgKV01FRFVvdnVlc01XOEJ4MTR  
L...  
QT1ZV1pKKz1qRkhFUTdBZXB3S1RtMEVraz0KLS0tLS1FTkQgUUFJJVkfURSBRLRVktLS0tLQo=
```

base64-encoded TLS Certificate:

```
LS0tLS1CRUdJTiBDRVJUSUZJQ0FURSB0tLS0tCk1JSUdGRENDQ1B5Z0F3SUJBZ01TQS9YVFJkaUpXOXMzb1  
dZZlN5aORpZ1h5TUEwR0NTcUdTSWlZRFFFQkN3VUEKTURNEN6QUUpCZ05WQkFZVEFsV1RNU113RkFZRFRZ  
UUtFdzFNW1hRbm  
...  
1ND0KLS0tLS1FTkQgQ0VSVE1GSUNBVEUtLS0tLQ==
```

Note: The PEM certificate and key must be further encoded using base64 as input to the installation screen. To encode a PEM certificate (e.g., cert.pem) to base64 on both Linux and Windows, you can use the following commands.

Linux: You can use the base64 command to read the cert.pem file, encodes it in base64, and saves the output to cert_base64.txt.

```
base64 cert.pem > cert_base64.txt
```

Windows: On Windows, you can use the certutil command to encode the cert.pem file in base64 and saves the result to cert_base64.txt.

```
certutil -encode cert.pem cert_base64.txt
```

Prerequisite: You should prepare these base64 encoded files before the start of the installation.

Press Next at the bottom of the page.

After you press Next, this page will be shown. When you are ready, scroll down and press Create.

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

luc.clement@k2view.com
K2VIEW (R&D)

Home > Marketplace >

Create K2view site deployment

activities. Microsoft does not provide rights for third-party offerings. See the [Azure Marketplace Terms](#) for additional details.

Name

Luc Clement

Preferred e-mail address

luc.clement@k2view.com

Preferred phone number

Basics

Subscription

K2view PAYG Marketing

Resource group

acm-k2view-rg

Region

East US 2

Application Name

k2view360

Managed Resource Group Name

mrg-k2view-azure-application

K2view Installation settings

Installation - User Assigned Identity Na...

k2view-identity

K2view Application settings

General - AKS Cluster Name

k2v-aks

General - Environment tag

Dev

General - Project tag

K2view

General - Domain ingress

acm.cloud.k2view.com

General - K2view cloud mailbox ID

d969cfcc-8346-4645-b60e-18fa0b334de8

K2view TLS Key

K2view TLS Certificate

Previous

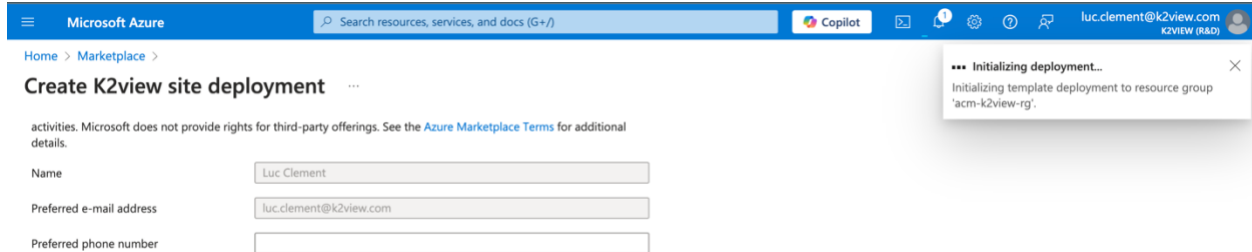
Next

Create

Give feedback

The Installation Process Begins

It will take approximately 15 minutes to create all resources in the **Azure Application Resource Group** And **Azure Managed Application Resource Group** you selected on the first page of the installation wizard.



Microsoft Azure

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Copilot

luc.clement@k2view.com K2VIEW (RAD)

Home > Marketplace >

Create K2view site deployment

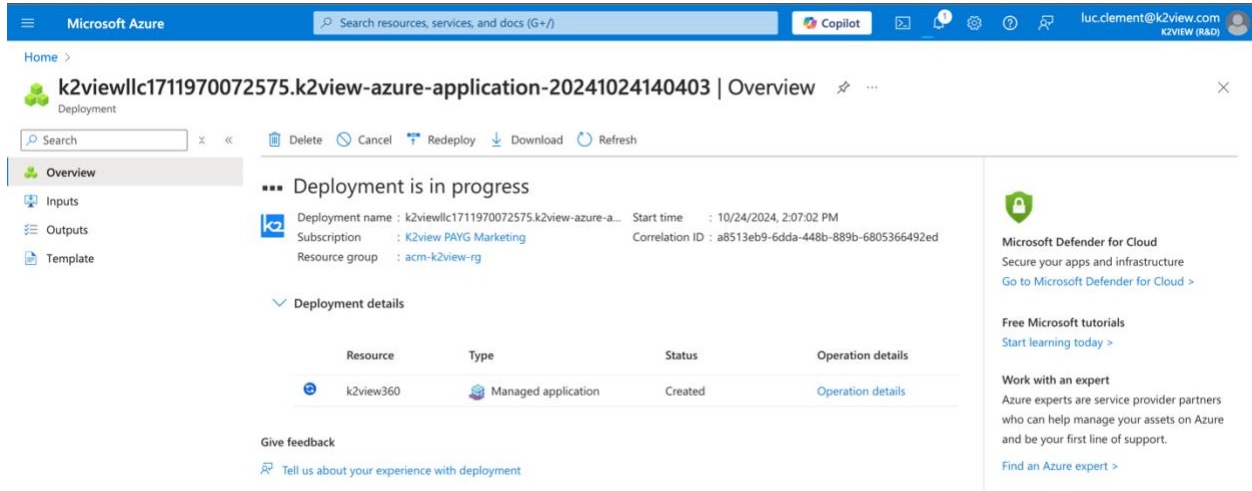
activities. Microsoft does not provide rights for third-party offerings. See the [Azure Marketplace Terms](#) for additional details.

Name:

Preferred e-mail address:

Preferred phone number:

*** Initializing deployment...
Initializing template deployment to resource group 'acm-k2view-rg'.



Microsoft Azure

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luc.clement@k2view.com K2VIEW (RAD)

Home >

k2viewllc1711970072575.k2view-azure-application-20241024140403 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

Deployment is in progress

Deployment name : k2viewllc1711970072575.k2view-azure-a... Start time : 10/24/2024, 2:07:02 PM
Subscription : K2view PAYG Marketing Correlation ID : a8513eb9-6dda-448b-889b-6805366492ed
Resource group : acm-k2view-rg

Deployment details

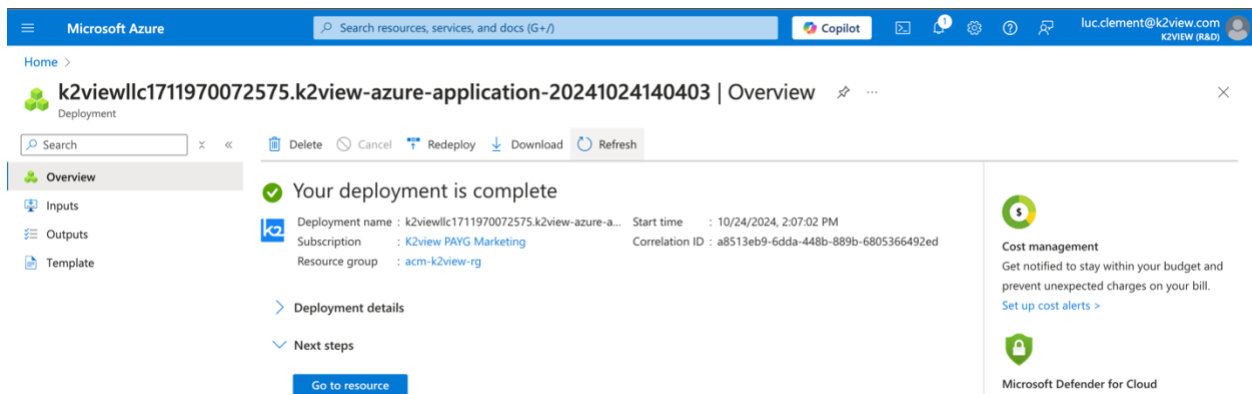
Resource	Type	Status	Operation details
k2view360	Managed application	Created	Operation details

Give feedback
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[Go to Microsoft Defender for Cloud >](#)

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[Find an Azure expert >](#)



Microsoft Azure

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Copilot

luc.clement@k2view.com K2VIEW (RAD)

Home >

k2viewllc1711970072575.k2view-azure-application-20241024140403 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

Your deployment is complete

Deployment name : k2viewllc1711970072575.k2view-azure-a... Start time : 10/24/2024, 2:07:02 PM
Subscription : K2view PAYG Marketing Correlation ID : a8513eb9-6dda-448b-889b-6805366492ed
Resource group : acm-k2view-rg

Deployment details

Next steps

[Go to resource](#)

Cost management
Get notified to stay within your budget and prevent unexpected charges on your bill.
[Set up cost alerts >](#)

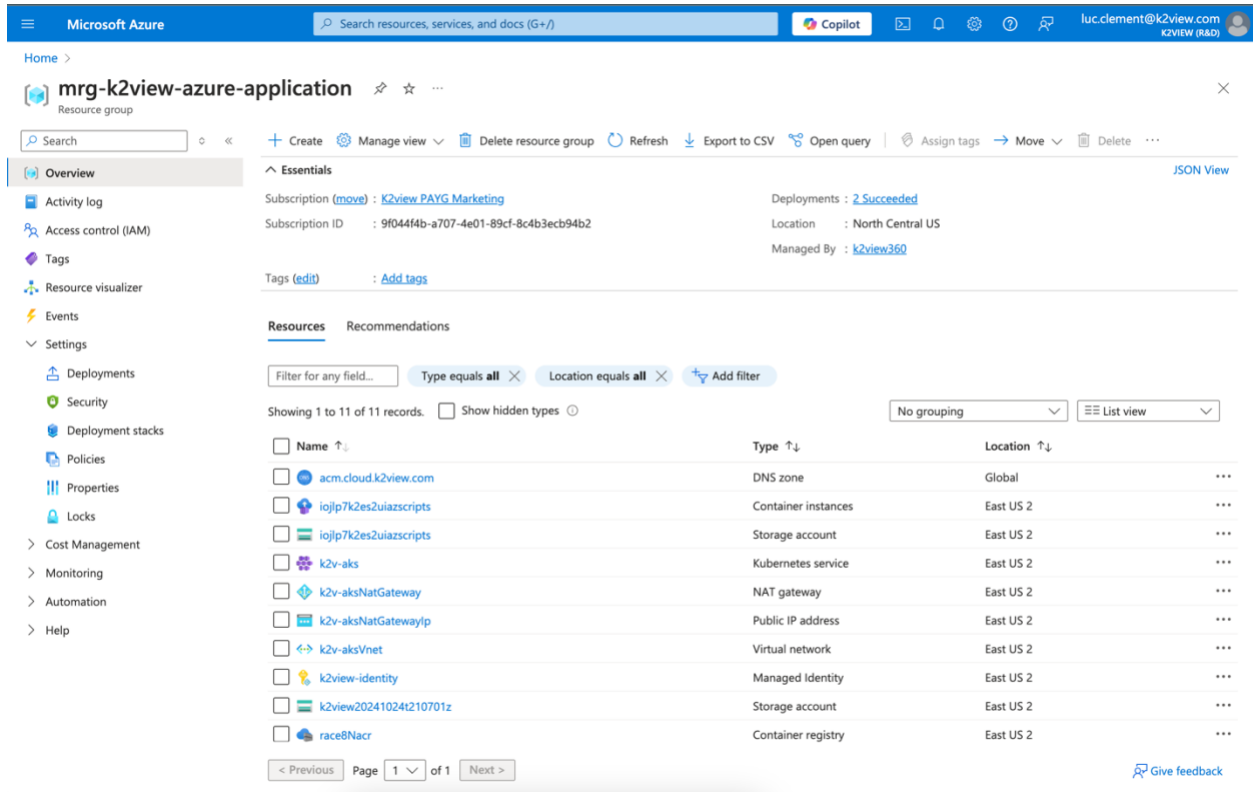
Microsoft Defender for Cloud
Secure your apps and infrastructure
[Go to Microsoft Defender for Cloud >](#)

The Installation has been Completed.

You will find the following created resources by navigating to the newly created resource group.

A Terraform module is used behind the scenes to take these input parameters to deploy K2View Fabric services in your environment. For more information on the details and configurations performed by the Azure Application installer, you can review the Terraform module available at:

<https://github.com/k2view/blueprints/tree/main/Azure/terraform/AKS>.

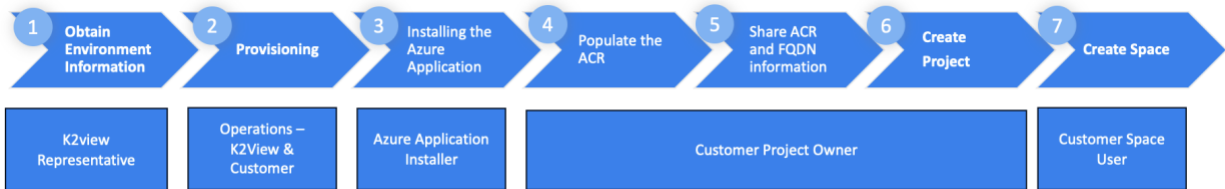


The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and user information. The main content area displays the 'mrg-k2view-azure-application' resource group. The 'Resources' tab is selected, showing a list of 11 resources. The resources are as follows:

Name	Type	Location
acm.cloud.k2view.com	DNS zone	Global
iojlp7k2es2uiazscripts	Container instances	East US 2
iojlp7k2es2uiazscripts	Storage account	East US 2
k2v-aks	Kubernetes service	East US 2
k2v-aksNatGateway	NAT gateway	East US 2
k2v-aksNatGatewayip	Public IP address	East US 2
k2v-aksVnet	Virtual network	East US 2
k2view-identity	Managed identity	East US 2
k2view20241024t210701z	Storage account	East US 2
race8Nacr	Container registry	East US 2

You are now ready to proceed to the next step: populate the Azure Container Registry created for you with K2view service Docker images.

4. Populating Your Azure Container Registry



To populate your Azure Container Repository, you will need

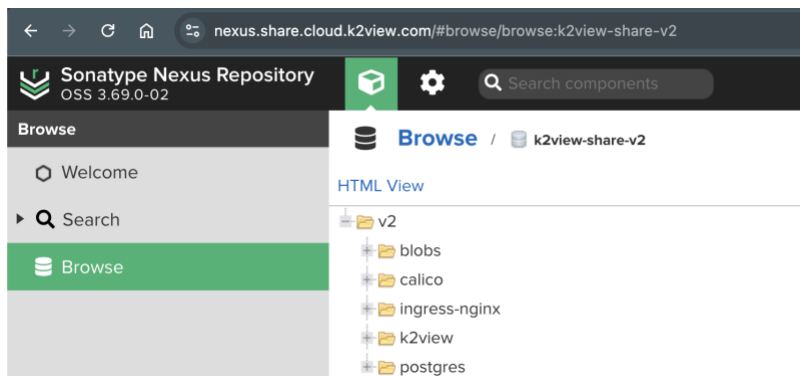
1. K2view Nexus Repository Account supplied to you by K2view
2. Obtained the list of Docker images to pull from Nexus and push to in consultation with K2view
3. Docker installed on the computer used to pull and push Docker images to your ACR

The [Planning](#) section describes what is necessary to be ready to perform this step.

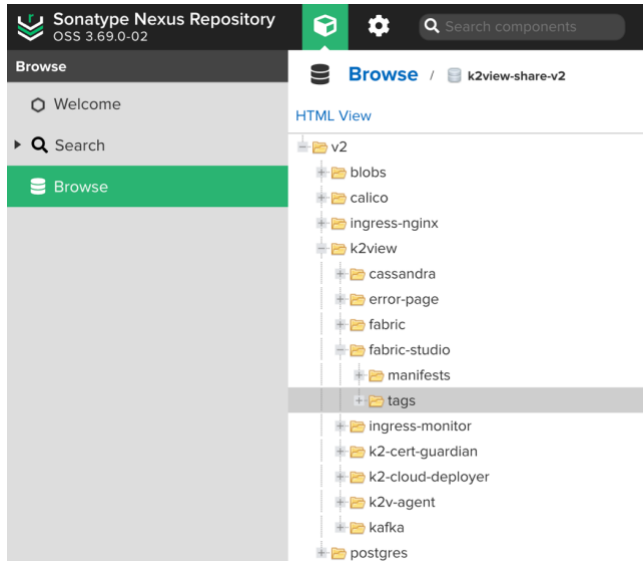
4.1. Getting Acquainted with the K2view Nexus Repository

To acquaint yourself of the K2view Docker images available to you can you login to the K2view Nexus web application. Here you can find the versions available to you. Please consult with your K2view representative which images and versions you should use. We recommend using the latest version of the K2view Docker images.

Login and navigate to <https://nexus.share.cloud.k2view.com/#browse/browse:k2view-share-v2> to view available images



For example, you can find the latest version of K2view's Fabric Studio by expanding the tags folder.



4.2. Commands to Pull Containers from the K2view Nexus Repository

The list of K2view service Docker images will be discussed and recommended to you in consultation with your K2view representative.

For the purpose of these instructions we will “pull” from the K2view Nexus repository three images of a specific version: fabric-studio, fabric, and the k2-cloud-deployer Docker images.

Docker tools need to be installed on the computer you are performing this installation. You can download these from <https://docker.com>.

First Login

Command:

```
docker login -u [your K2view Nexus username] https://docker.share.cloud.k2view.com
```

Second, pull the required images. In this example we pull three.

Commands: In succession, please run the following three commands. Keep in mind that the version and docker images needed may differ.

```
docker pull docker.share.cloud.k2view.com/k2view/fabric-studio:8.1.0_199
```

```
docker pull docker.share.cloud.k2view.com/k2view/fabric:8.1.0_199
```

```
docker pull docker.share.cloud.k2view.com/k2view/k2-cloud-deployer:1.8.0
```

4.3. Commands to Push Containers to the Azure Container Registry

Now that you have “pulled” images from the K2view Nexus repository, you are now ready to “push” these to the Azure Container Registry (ACR) created for you. If you wish to use a preexisting ACR or OCI-compliant container registry, please also consult the [Selecting a Different Azure Container Registry or OCI-Compliant Registry](#) section.

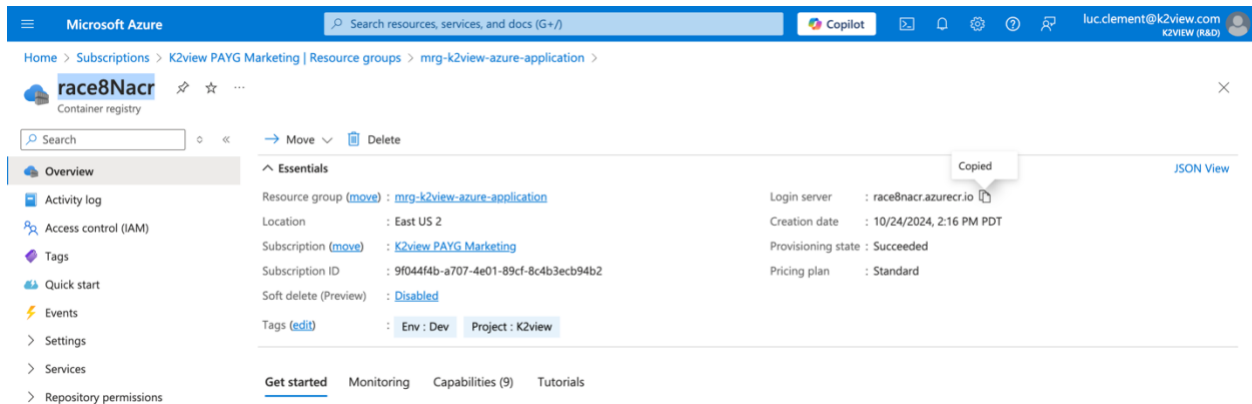
First, obtain the credentials required to perform a Docker Push to your ACR

Installing the K2view Azure Application created an ACR within the Azure Managed Application Resource Group. You can obtain the credential associated with it by navigating you the resource group and selecting the ACR created for you.

In this example, the Managed Application Resource Group is named mgr-k2view-azure-application, and the created ACR is named rac8Nacr (a random name).

When you navigate to this ACR location, please

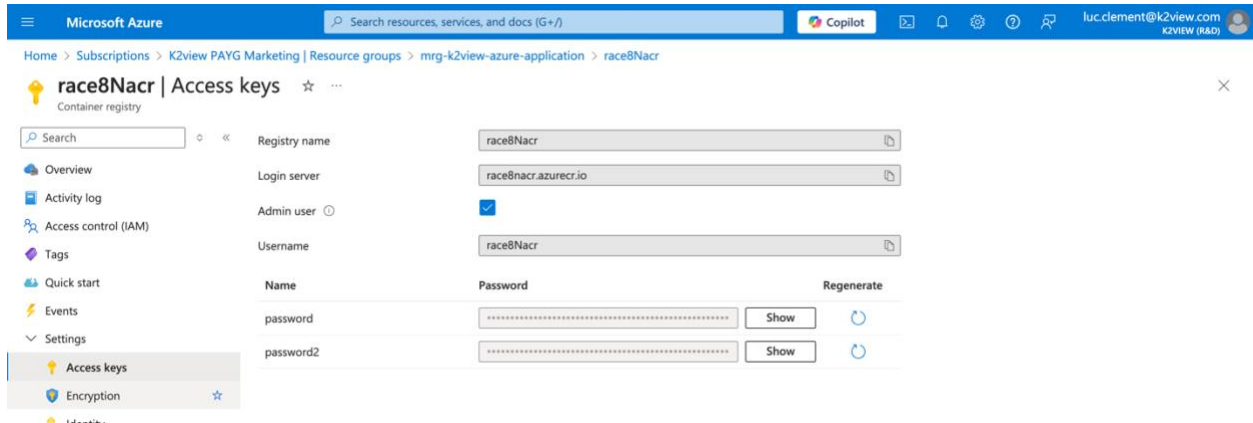
1. Get the URL of the login server, which is shown here. Copy the URL which will be required for your Docker login and push commands.



The screenshot shows the Microsoft Azure portal interface. The breadcrumb navigation at the top indicates the path: Home > Subscriptions > K2view PAYG Marketing | Resource groups > mgr-k2view-azure-application > race8Nacr. The 'race8Nacr' resource is identified as a 'Container registry'. On the left sidebar, the 'Overview' tab is selected. The main content area displays the 'Essentials' section for the ACR. Key information includes: Resource group (mgr-k2view-azure-application), Location (East US 2), Subscription (K2view PAYG Marketing), Subscription ID (9f044f4b-a707-4e01-89cf-8c4b3ecb94b2), and Soft delete (Preview) (Disabled). The 'Tags' section shows 'Env : Dev' and 'Project : K2view'. On the right, the 'Login server' is listed as 'race8nacr.azurecr.io', with a 'Copied' tooltip indicating it has been copied. Other details include 'Creation date : 10/24/2024, 2:16 PM PDT', 'Provisioning state : Succeeded', and 'Pricing plan : Standard'. At the bottom, there are tabs for 'Get started', 'Monitoring', 'Capabilities (9)', and 'Tutorials'.

2. Select Access Keys in the left navigation menu under Settings to obtain your credentials. From this page:
 - a. Copy the username
 - b. Use the Show button to obtain the password, and copy this password

This information will be required to perform the login to your ACR using a Docker command



Second, perform the Docker Push to your ACR

Use the Docker login command to login to your ACR

```
docker login -u [your ACR username] [the ACR Login server]
```

In our example we used: `docker login -u race8Nacr race8nacr.azurecr.io`

Third, tag each image you pulled from the K2view Nexus

Use of tags is for version control. Tagging Docker images allows you to maintain different versions of the same application, making it easier to roll back to a previous version if needed. This can be especially useful in cases where a new release introduces a bug or breaks compatibility with other components.

In our example, we used the following three commands, respectively, for the fabric-studio, fabric, and k2-cloud-deployer Docker images. Note that the highlighted **race8nacr** value is the name used in our example, you will need to use the one associated with your ACR.

```
docker tag docker.share.cloud.k2view.com/k2view/fabric-studio:8.1.0_199  
race8nacr.azurecr.io/k2view/fabric-studio:8.1.0_199
```

```
docker tag docker.share.cloud.k2view.com/k2view/fabric:8.1.0_199  
race8nacr.azurecr.io/k2view/fabric:8.1.0_199
```

```
docker tag docker.share.cloud.k2view.com/k2view/k2-cloud-deployer:8.1.0_199  
race8nacr.azurecr.io/k2view/k2-cloud-deployer:8.1.0_199
```

Fourth, push images to your ACR

Next, you need to push the Docker images you pulled from the K2view Nexus to your ACR using the command: `docker push [image location]`.

In our example, we used:

```
docker push race8nacr.azurecr.io/k2view/fabric-studio:8.1.0_199
```

```
docker push race8nacr.azurecr.io/k2view/fabric:8.1.0_199
```

```
docker push race8nacr.azurecr.io/k2view/k2-cloud-deployer:1.8.0
```

Fifth, please provide the locations you used to K2view

K2view needs these ACR locations of your images to finalize your site's configuration. Please note that you will share these locations with K2view as described in the [Sharing Information with K2view](#) section.

4.4. Selecting a Different Azure Container Registry or OCI-Compliant Registry

You must follow similar steps if you choose to use a pre-existing ACR or an OCI-compliant registry. You will also need to provide K2view with the locations of your registry.

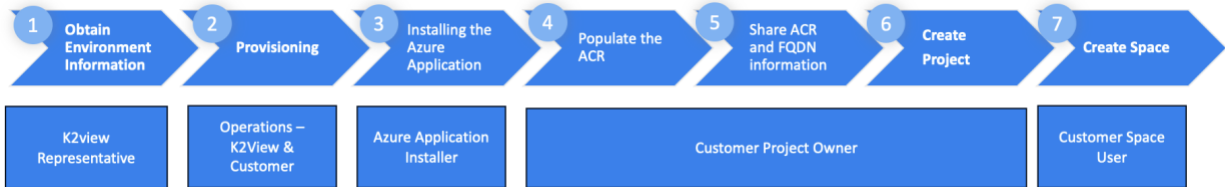
4.5. Sharing Information with K2view

The Customer Project Owner must share the following information to complete the configuration for the tenant's creation of Projects and Spaces.

- The fully qualified domain name (FQDN) used for your Domain Ingress.
 - For example, `acm.cloud.k2view.com` was used. You need to use a domain your organization owns.
- The list of Azure Container Registry (ACR) locations of your Docker images
 - E.g., the locations shown in the fourth step of the [Commands to Push Containers to the Azure Container Registry](#) section.
- The AKS Cluster name

This information may have been identified during pre-planning, but the specific information should be confirmed to avoid misconfiguration.

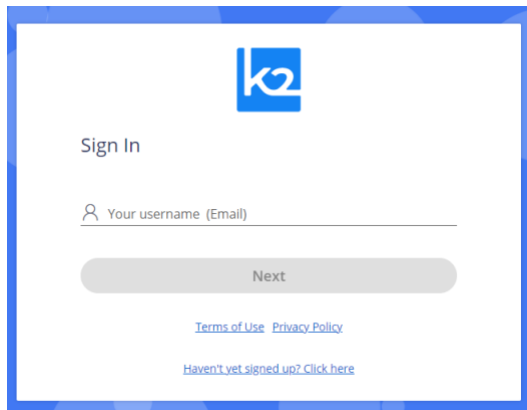
5. Creating Your Fabric Project



5.1. Sign In to K2cloud

To sign in, access cloud.k2view.com using Chrome, Microsoft Edge, or Firefox web browsers.

When prompted to sign in, enter your email address. This account needs to have been provisioned for you by K2view. Please refer to the [Prerequisites](#) section.



If provisioned by K2view, you will receive an access code via email you will then enter in the prompt window that is shown when you press Next.

You will be directed to the Spaces page, where you will eventually create spaces. First, you need to create a project.

5.2. Create the Project

You create a project by selecting the Projects top-level navigation tab and pressing “Create Project”. A project requires a name and an associated Git repository. You must provide Git-related information to create a project. To learn more about Git and how to obtain the Repository URL you will use, a branch, and your token, please review the [GitHub](#) section.

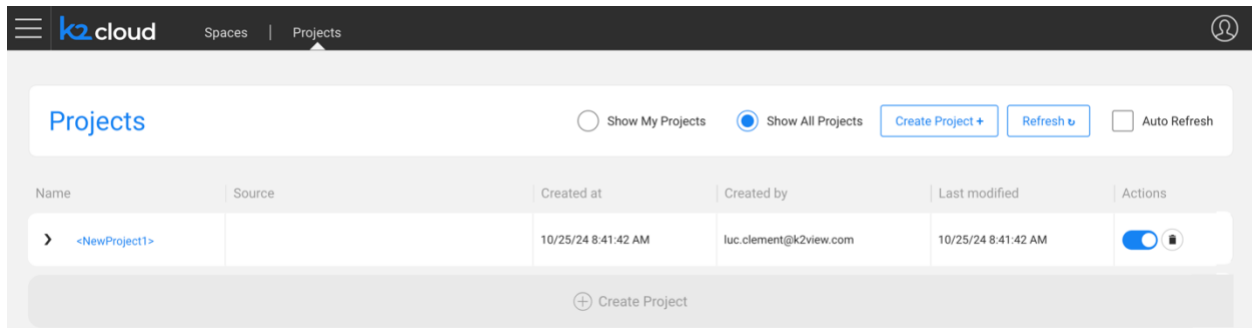
The Fabric spaces you implement are based on a “Space Profile”. You can configure more than one Space Profile per Project. If you need assistance, a K2view representative will assist you. Some of this information will have been shared with you during the [Planning](#) phase.

To create a Space Profile, you need to:

- Give it a name that will have significance to your users
- Select an Image Profile identified in consultation with K2view, and
- Define the site where this space will operate within. K2view will help you define this list.

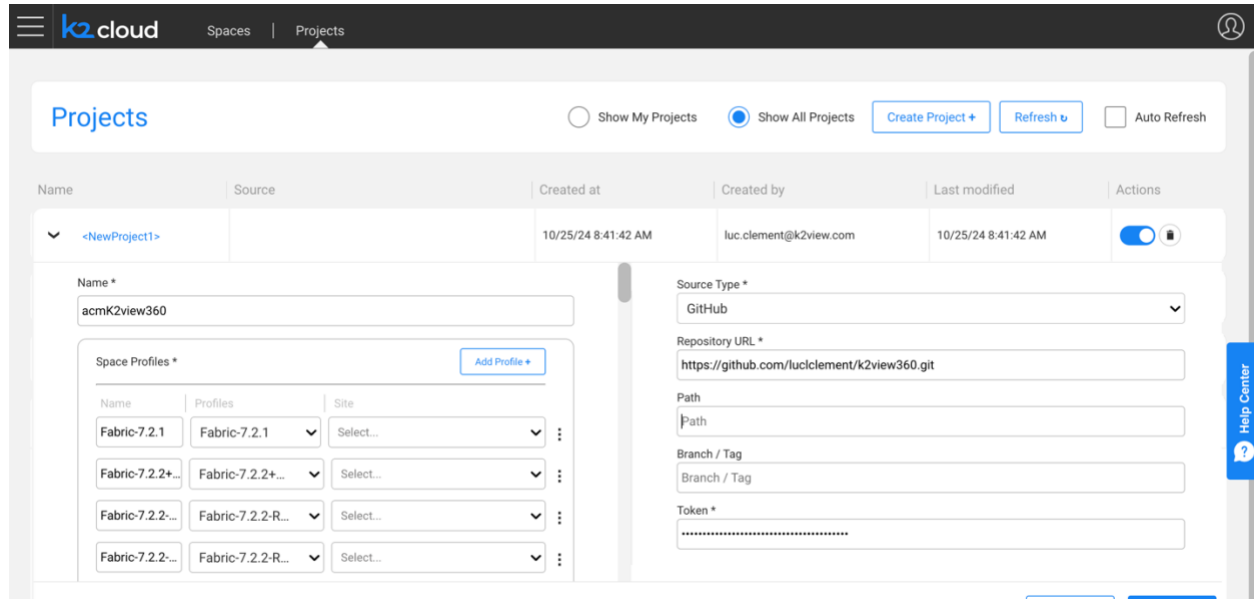
Creating a Project

After pressing Create Project, an empty project will be created.



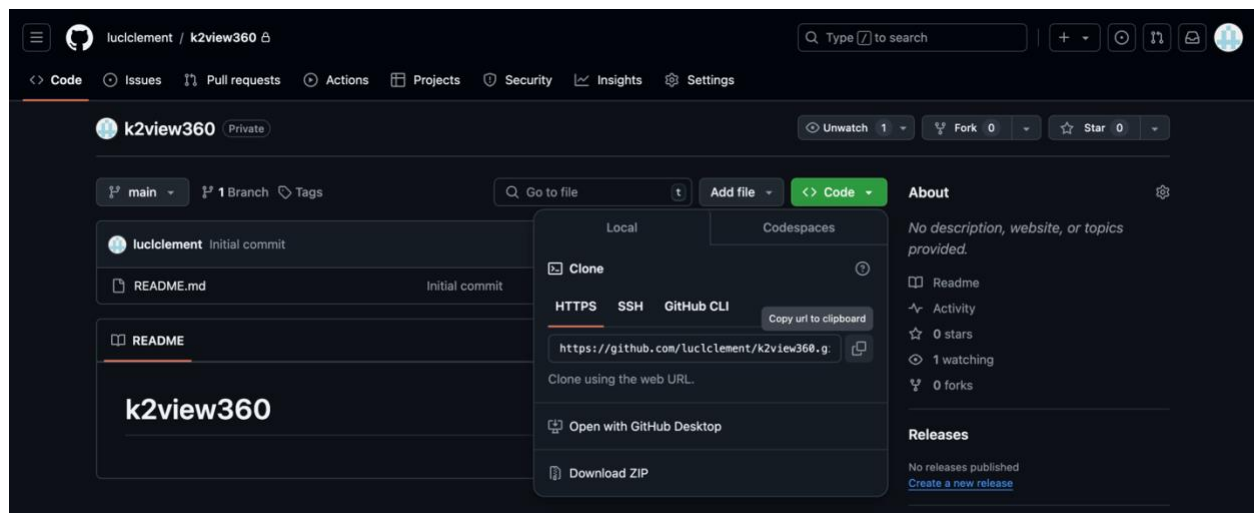
Provide the project with a name and select your space profiles. Again, K2view will have provided this.

You also need to provide the details of your Git source control system described in the [GitHub](#) section.

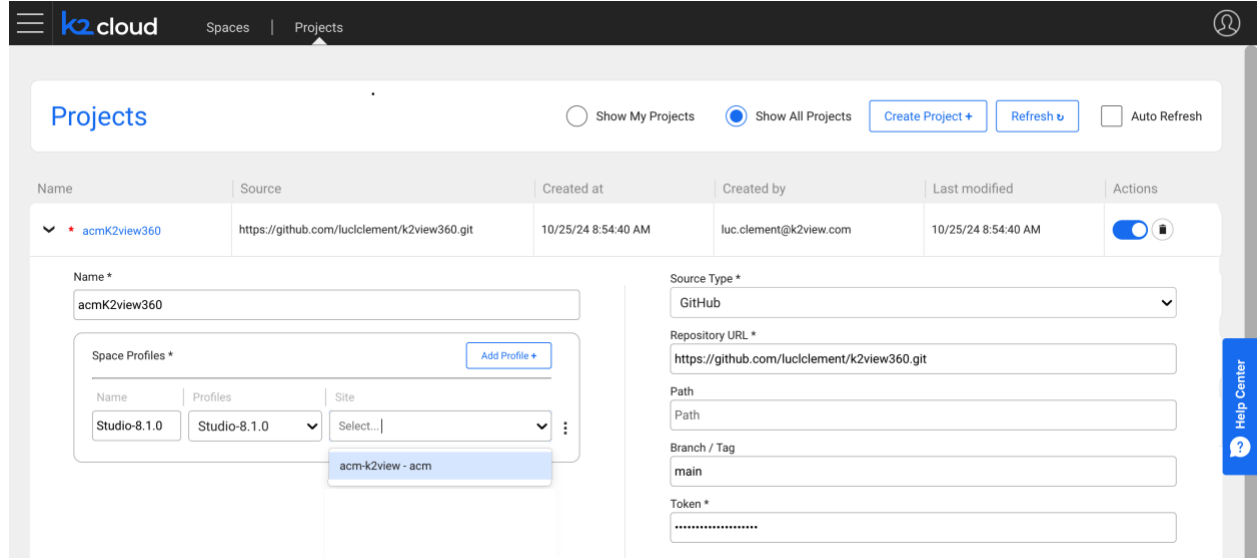


The screenshot shows the 'Projects' configuration page in the K2view cloud interface. The page has a header with 'k2cloud' and navigation tabs for 'Spaces' and 'Projects'. Below the header, there are buttons for 'Show My Projects', 'Show All Projects' (selected), 'Create Project +', 'Refresh', and 'Auto Refresh'. A table lists projects with columns: Name, Source, Created at, Created by, Last modified, and Actions. The first project is '<NewProject1>'. Below the table, the configuration form is visible. It includes a 'Name *' field with 'acmK2view360', a 'Source Type *' dropdown set to 'GitHub', a 'Repository URL *' field with 'https://github.com/luccllement/k2view360.git', a 'Path' field, a 'Branch / Tag' field, and a 'Token *' field. There is also a 'Space Profiles *' section with a table of profiles and a 'Add Profile +' button.

For example, you can obtain the URL from your Git repository, as shown here in our example.



And, finally, please select the site created from the site's dropdown.



The screenshot shows the K2view cloud interface. At the top, there's a navigation bar with 'k2cloud', 'Spaces', and 'Projects'. The 'Projects' section is active, showing a table of projects. Below the table, there's a form to create a new project. The form includes fields for Name, Source Type, Repository URL, Path, Branch / Tag, and Token. A dropdown menu for 'Site' is open, showing a list of sites, with 'acm-k2view - acm' selected.

Name	Source	Created at	Created by	Last modified	Actions
acmK2view360	https://github.com/luccllement/k2view360.git	10/25/24 8:54:40 AM	luc.clement@k2view.com	10/25/24 8:54:40 AM	

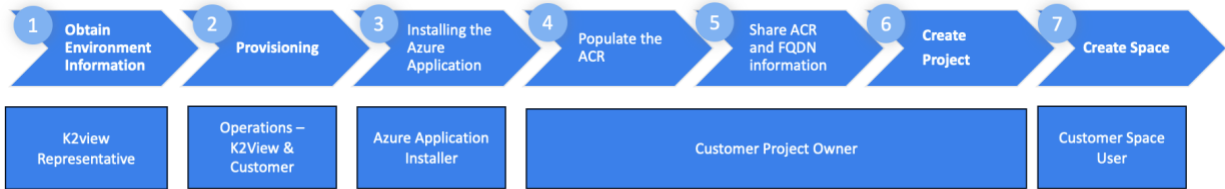
Form Fields:

- Name ***: acmK2view360
- Source Type ***: GitHub
- Repository URL ***: https://github.com/luccllement/k2view360.git
- Path**: Path
- Branch / Tag**: main
- Token ***:
- Space Profiles ***:

Name	Profiles	Site
Studio-8.1.0	Studio-8.1.0	Select...

 (Dropdown menu open showing: acm-k2view - acm)

6. Creating Your First Fabric Space



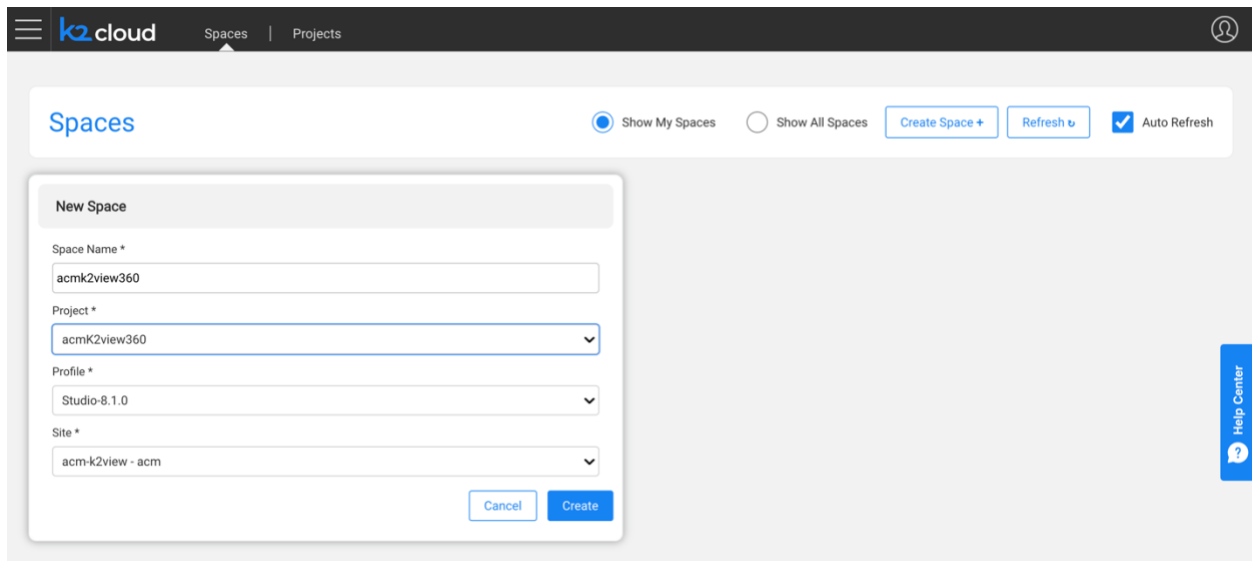
After your Project is created, you can create a Fabric Space. Here, we show an example of an existing space and a new space being defined.

To create a space

- Give it a name,
- Select the project you created,
- Select your intended profile, one recommended by K2view given your intended purpose, and
- Select site.

If you need to install a K2view extension such as TDM, you must go to your space and install the extension from K2exchange.

When ready, press Create to trigger the space's creation.

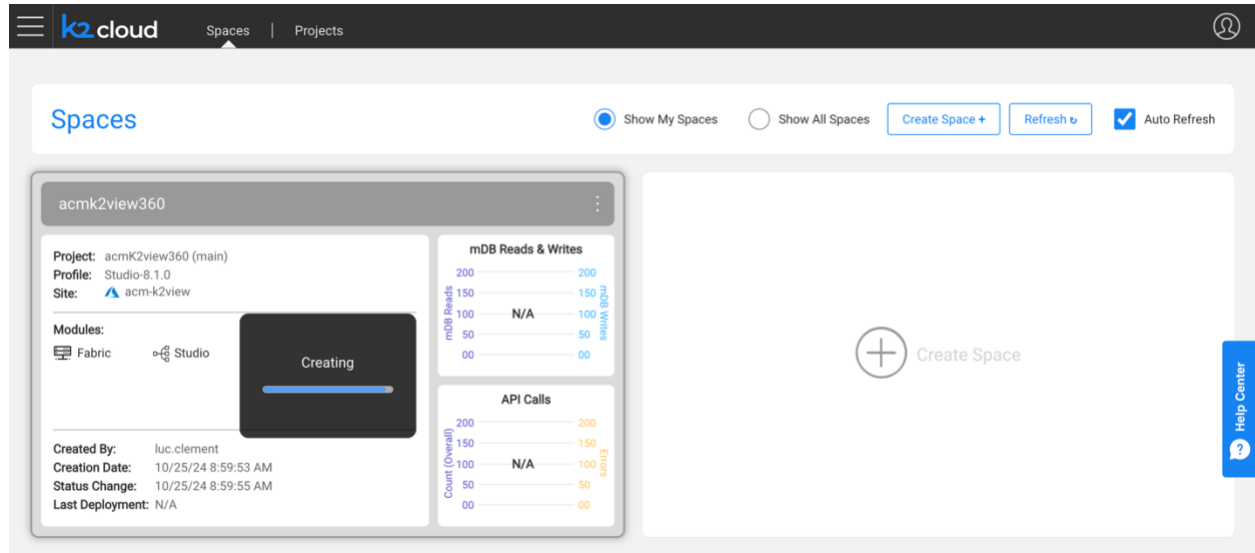


The screenshot shows the K2cloud interface with the 'Spaces' tab selected. A 'New Space' modal is open, allowing the user to create a new space. The modal contains the following fields:

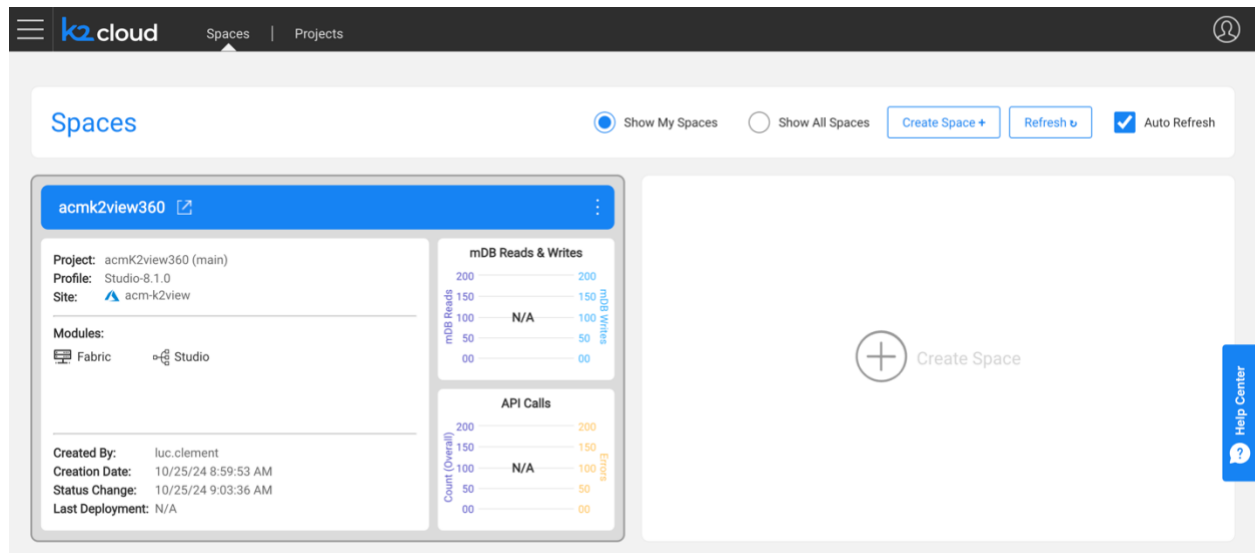
- Space Name ***: acmk2view360
- Project ***: acmk2view360 (selected from a dropdown)
- Profile ***: Studio-8.1.0 (selected from a dropdown)
- Site ***: acm-k2view - acm (selected from a dropdown)

At the bottom of the modal are 'Cancel' and 'Create' buttons. In the background, the 'Spaces' page shows a list of spaces and a 'Create Space +' button. The top navigation bar includes the K2cloud logo and a user profile icon.

The Space creation process begins and may take a few minutes to complete.



Once complete your site is displayed to you.



You have successfully created your Project and Space and are ready to begin working with K2view.

7. Validation

You are now ready to proceed with validating your space. Doing so is performed by selecting the top right arrow icon (see next section).

This will cause the browser to connect to the Space you created over HTTPS to the environment you created. The TLS certificate you assigned will be used for this connection.

DNS

During the installation, a DNS zone was created. This zone may take time to propagate into your environment and may not be available until it does.

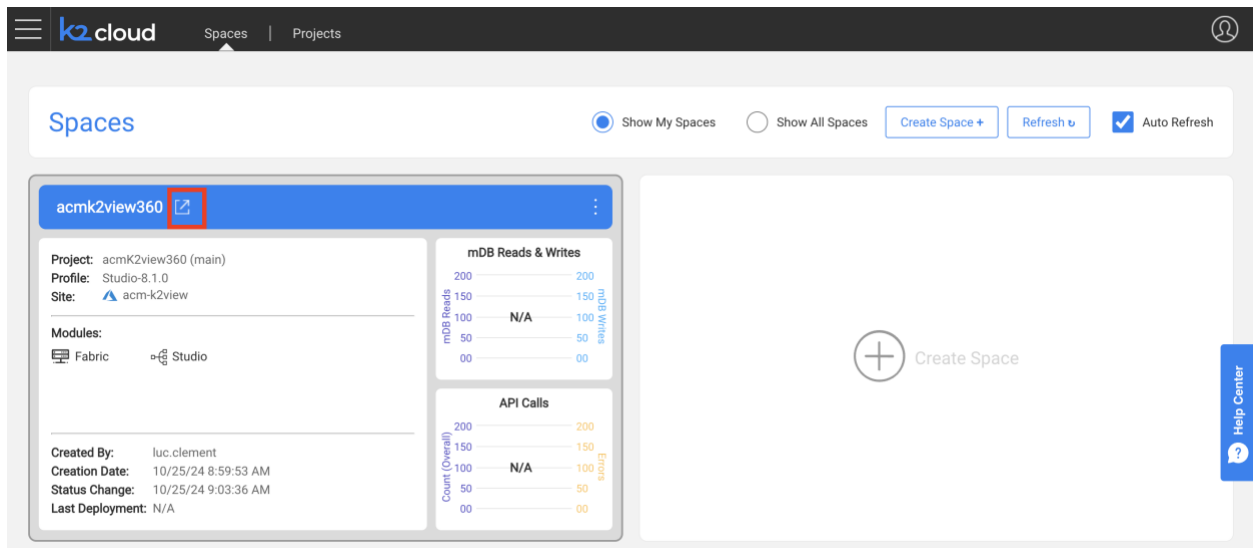
You should note that when creating a namespace, its name is associated as a subdomain to this domain name in the Ingress controller. For example, if the domain is "k2dev.company.com" and a created namespace is "test", then the URL of this namespace that users shall access will be "test.k2dev.company.com".

In our example, we will connect to a space accessible at acmk2view360-k2v-rnd.acm.cloud.k2view.com, a publicly resolvable domain.

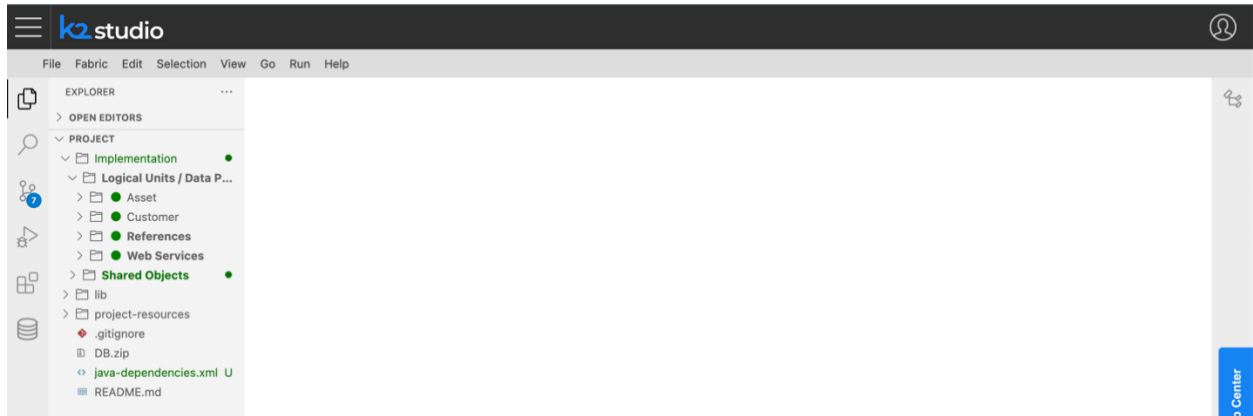
Note: Some organizations may not allow the use of DNS records created in your installation's Public DNS zone to propagate your organization's DNS. If that's the case, you must provide the address information to your organization. Refer to the [DNS zone](#) section for this.

7.1. Enter The Fabric Space

To enter your space, select the top left corner to launch the workspace.



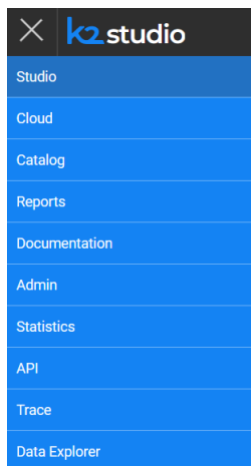
The **Studio** web application will be displayed.



Then, navigate to the various applications' context menus and confirm that each web application appears functional by selecting the “hamburger” menu item on the top left of the page.



The following is the list of web applications you should expect to see.



8. Completion

This completes the end of your installation and configuration of the K2view Azure Application, the creation of your K2view Project and the creation of your first K2view Project.

We invite you to visit our Knowledge Base at <https://support.k2view.com/knowledge-base.html>, where you can find documentation and demo projects, and sign up for the K2view Academy Learning Center at <https://academy.k2view.com>. For access, please talk to your K2view representative.

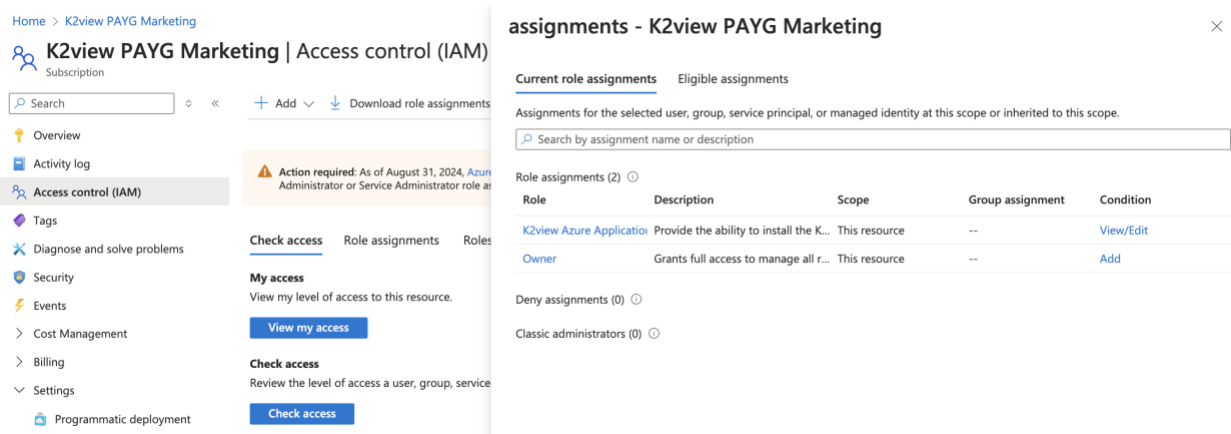
9. Instructions

9.1. Permissions and Custom Role Creation

The individual performing the K2view Azure Application installation needs a Subscription-level Owner role and a custom role that grants specific permissions to perform the installation. This user needs:

- A Subscription Owner Role (i.e., the Owner role within the Subscription). This is required to be able to create the resource groups the installer will generate.
- And have an associated [Custom Role](#) that grants the user these permissions.
 1. "Microsoft.Authorization/roleAssignments/write",
 2. "Microsoft.Authorization/roleAssignments/read",
 3. "Microsoft.Authorization/roleAssignments/delete"
- A role assignment of this custom role needs to be made for this individual

Once assigned the role will appear for the user. In this example the custom role is named "K2view Azure Application Installation".



Home > K2view PAYG Marketing

K2view PAYG Marketing | Access control (IAM)

Subscription

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Security

Events

Cost Management

Billing

Settings

Programmatic deployment

Action required: As of August 31, 2024, Azure Administrator or Service Administrator role as

Check access

Role assignments

Roles

My access

View my level of access to this resource.

View my access

Check access

Review the level of access a user, group, service

Check access

assignments - K2view PAYG Marketing

Current role assignments

Eligible assignments

Assignments for the selected user, group, service principal, or managed identity at this scope or inherited to this scope.

Search by assignment name or description

Role	Description	Scope	Group assignment	Condition
K2view Azure Application	Provide the ability to install the K...	This resource	--	View/Edit
Owner	Grants full access to manage all r...	This resource	--	Add

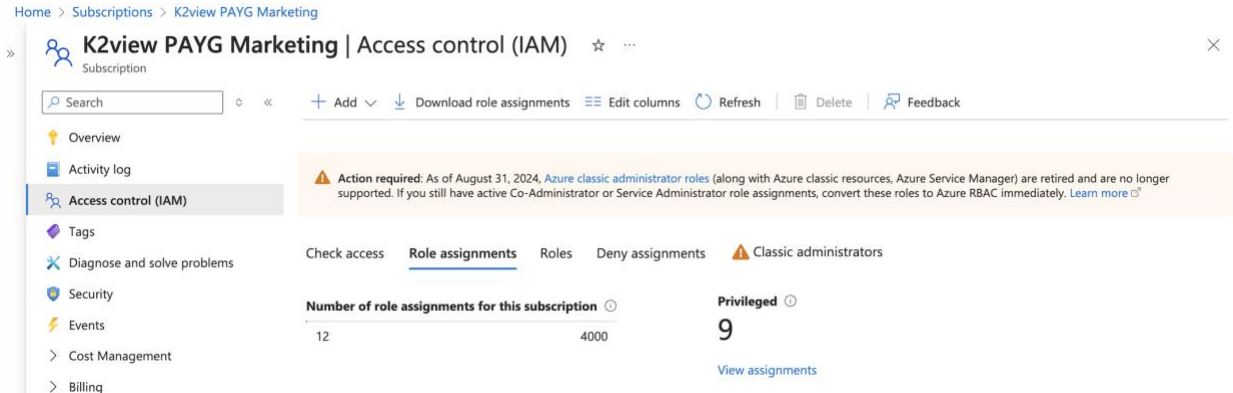
Deny assignments (0)

Classic administrators (0)

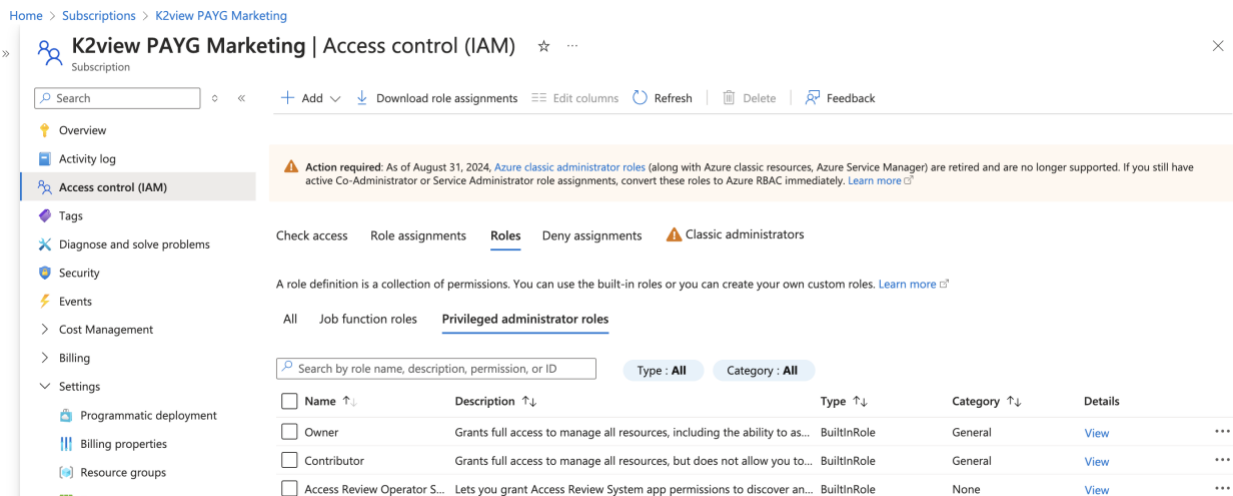
This section describes the steps to create a custom role with the associated permissions.

Within the Subscription's Access Control Section

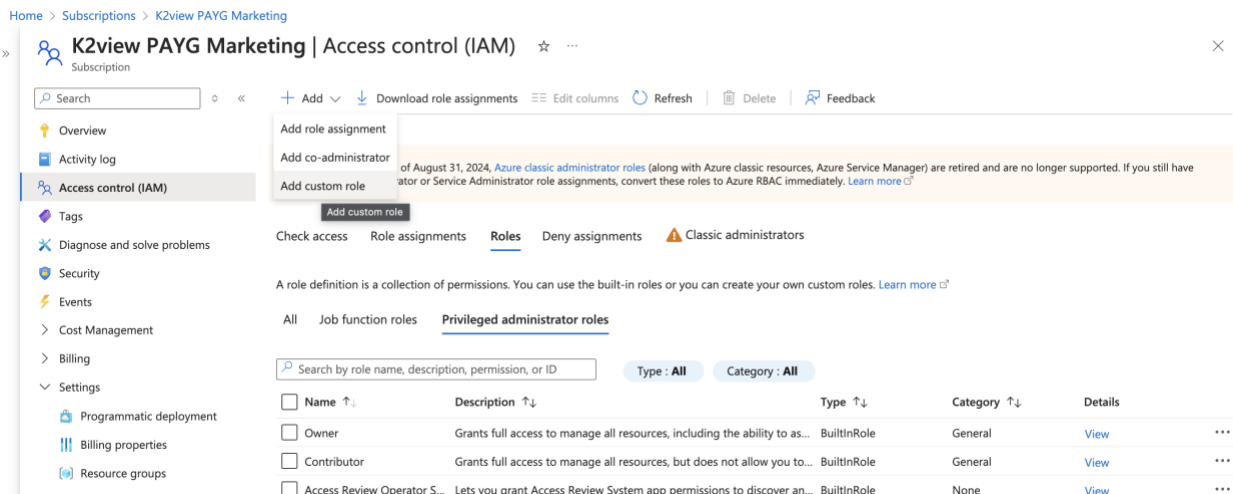
Navigate to the Subscription's Access Control (IAM) section



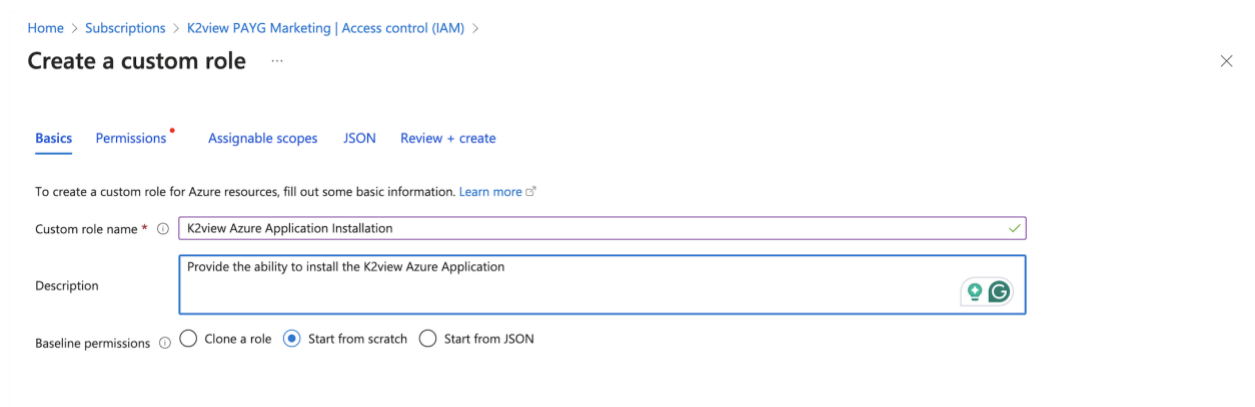
Select the Roles tab



Select the Add drop down on the top left of the panel



Using the Create a custom role page enter a name that is significant for your organization. In our example we used K2view Azure Application Installation as the name.



Home > Subscriptions > K2view PAYG Marketing | Access control (IAM) >

Create a custom role

Basics Permissions Assignable scopes JSON Review + create

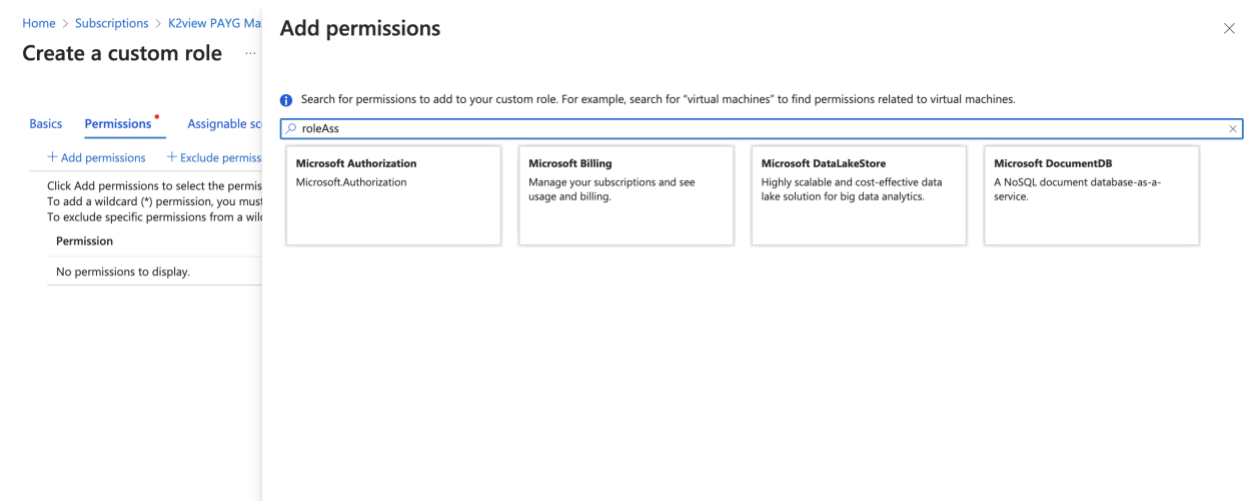
To create a custom role for Azure resources, fill out some basic information. [Learn more](#)

Custom role name * K2view Azure Application Installation ✓

Description Provide the ability to install the K2view Azure Application

Baseline permissions ☐ Clone a role ☒ Start from scratch ☐ Start from JSON

Press next and select Add permission ... search for roleAss. This will display the “Microsoft Authorization” panel.



Home > Subscriptions > K2view PAYG Marketing | Access control (IAM) >

Create a custom role

Basics Permissions Assignable scopes

+ Add permissions + Exclude permissions

Click Add permissions to select the permissions to add to your custom role. To add a wildcard (*) permission, you must add a specific permission from a wildcard permission.

Permission

No permissions to display.

Add permissions

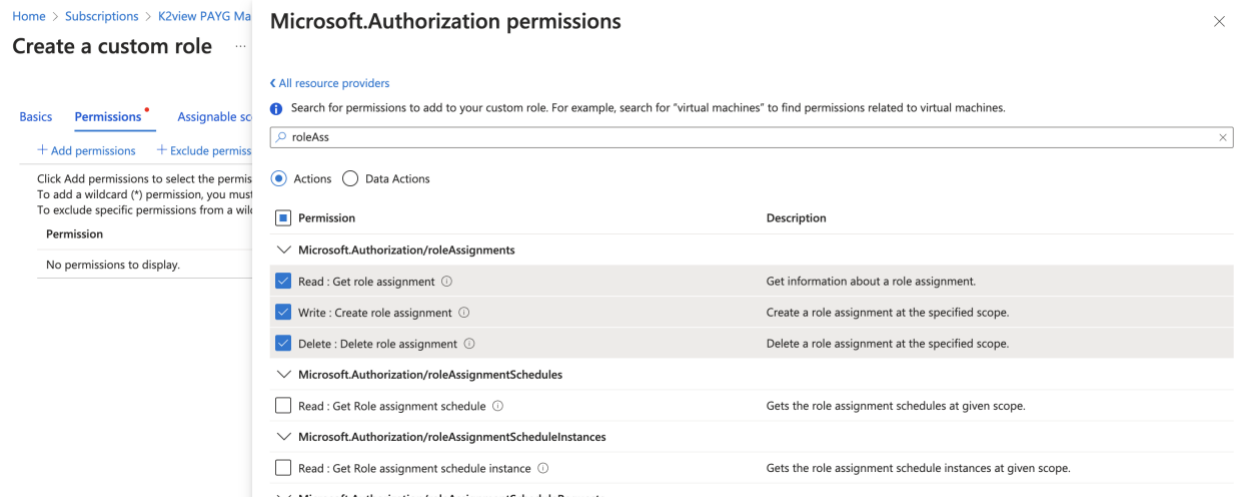
Search for permissions to add to your custom role. For example, search for "virtual machines" to find permissions related to virtual machines.

roleAss

Microsoft Authorization Microsoft.Authorization	Microsoft Billing Manage your subscriptions and see usage and billing.	Microsoft DataLakeStore Highly scalable and cost-effective data lake solution for big data analytics.	Microsoft DocumentDB A NoSQL document database-as-a-service.
---	--	---	--

Please select the “Microsoft Authorization” panel.

In the Microsoft.Authorization/roleAssignments section, please select the Read, Write and Delete assignments. Microsoft will require these three assignments to allow the user to perform the installation of the K2view Azure Application and the components it needs to install in two Azure resource groups: one for the K2view Application and the second one for the K2view Fabric resources.



Home > Subscriptions > K2view PAYG Marketing > Create a custom role

Basics Permissions Assignable scope

+ Add permissions + Exclude permissions

Click Add permissions to select the permissions. To add a wildcard (*) permission, you must select specific permissions from a list.

Permission

No permissions to display.

Microsoft.Authorization permissions

< All resource providers

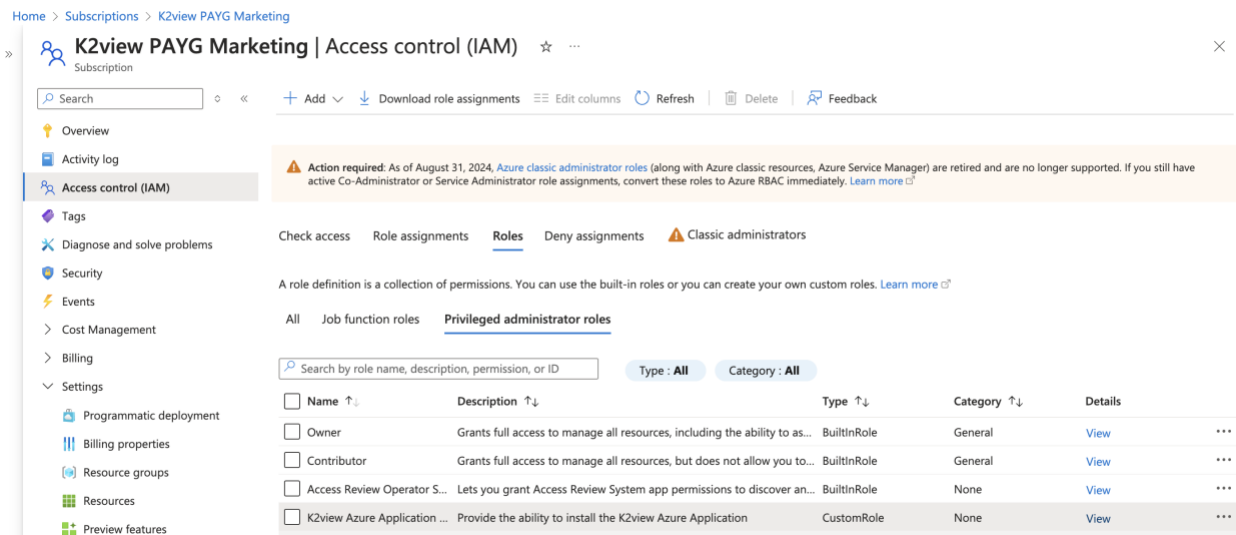
Search for permissions to add to your custom role. For example, search for "virtual machines" to find permissions related to virtual machines.

roleAss

Actions Data Actions

Permission	Description
☒ Read : Get role assignment	Get information about a role assignment.
☒ Write : Create role assignment	Create a role assignment at the specified scope.
☒ Delete : Delete role assignment	Delete a role assignment at the specified scope.
☐ Read : Get Role assignment schedule	Gets the role assignment schedules at given scope.
☐ Read : Get Role assignment schedule instance	Gets the role assignment schedule instances at given scope.

You should now be able to see new custom role in the “Privileged administrator roles” section.



Home > Subscriptions > K2view PAYG Marketing > K2view PAYG Marketing | Access control (IAM)

Subscription

Search

+ Add Download role assignments Edit columns Refresh Delete Feedback

Overview Activity log Access control (IAM) Tags Diagnose and solve problems Security Events Cost Management Billing Settings

Programmatic deployment Billing properties Resource groups Resources Preview features

Action required: As of August 31, 2024, Azure classic administrator roles (along with Azure classic resources, Azure Service Manager) are retired and are no longer supported. If you still have active Co-Administrator or Service Administrator role assignments, convert these roles to Azure RBAC immediately. [Learn more](#)

Check access Role assignments Roles Deny assignments Classic administrators

A role definition is a collection of permissions. You can use the built-in roles or you can create your own custom roles. [Learn more](#)

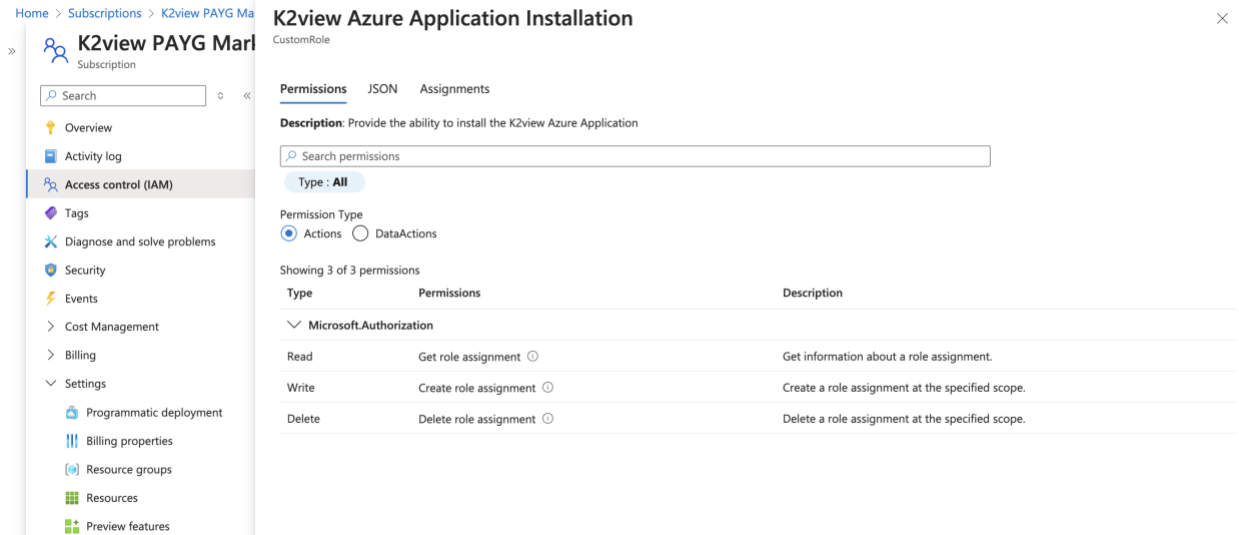
All Job function roles Privileged administrator roles

Search by role name, description, permission, or ID

Type: All Category: All

Name	Description	Type	Category	Details
Owner	Grants full access to manage all resources, including the ability to as...	BuiltInRole	General	View
Contributor	Grants full access to manage all resources, but does not allow you to...	BuiltInRole	General	View
Access Review Operator S...	Lets you grant Access Review System app permissions to discover an...	BuiltInRole	None	View
K2view Azure Application ...	Provide the ability to install the K2view Azure Application	CustomRole	None	View

Exploring this custom role further, you will see the assigned permissions.



K2view Azure Application Installation
CustomRole

Permissions JSON Assignments

Description: Provide the ability to install the K2view Azure Application

Search permissions

Type: **All**

Permission Type
☒ Actions ☐ DataActions

Showing 3 of 3 permissions

Type	Permissions	Description
▼ Microsoft.Authorization		
Read	Get role assignment	Get information about a role assignment.
Write	Create role assignment	Create a role assignment at the specified scope.
Delete	Delete role assignment	Delete a role assignment at the specified scope.

Now, assign the role to the user performing the installation. Use the Assignment tab for this, and select “Add assignment”. The user performing this task needs the permissions to be able to make the assignment of roles and custom roles.



K2view Azure Application Installation
CustomRole

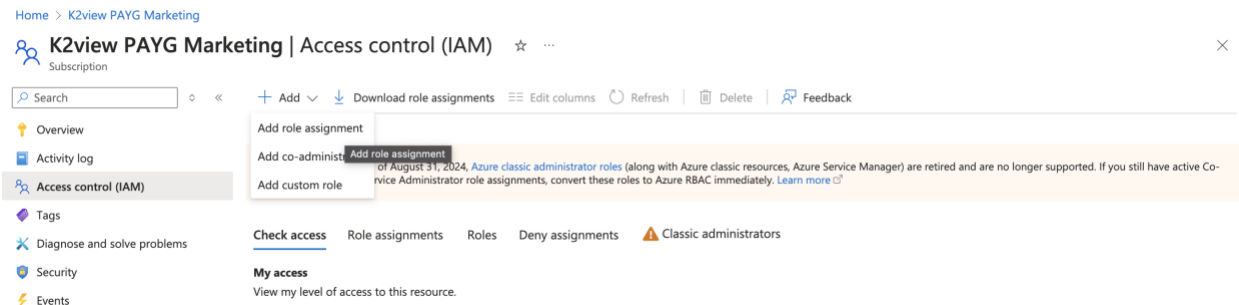
Permissions JSON **Assignments**

Search assignments by name

+ Add assignment

Name	Type	Scope
------	------	-------

Alternatively, you can use the Add role assignment drop down menu item in the Access control (IAM) panel.



Home > K2view PAYG Marketing

K2view PAYG Marketing | Access control (IAM) ☆ ...

Search

+ Add ▼ Download role assignments Edit columns Refresh Delete Feedback

Add role assignment

Add co-administrator **Add role assignment**

Add custom role

of August 31, 2024, [Azure classic administrator roles](#) (along with Azure classic resources, Azure Service Manager) are retired and are no longer supported. If you still have active Co-Service Administrator role assignments, convert these roles to Azure RBAC immediately. [Learn more](#)

Check access Role assignments Roles Deny assignments Classic administrators

My access
View my level of access to this resource.

Using the Add role assignment select the custom role and add the user to this role following these steps.

Home > Subscriptions > K2view PAYG Marketing | Access control (IAM) >

Add role assignment

Role Members Conditions Review + assign

A role definition is a collection of permissions. You can use the built-in roles or you can create your own custom roles. [Learn more](#)

Job function roles **Privileged administrator roles**

Grant privileged administrator access, such as the ability to assign roles to other users.

⚠ Can a job function role with less access be used instead?

🔍 Search by role name, description, permission, or ID

Type: All

Category: All

Name ↑↓	Description ↑↓	Type ↑↓	Category ↑↓	Details
Contributor	Grants full access to manage all resources, but does not allow you to assign roles in Azure RBA...	BuiltInRole	General	View
Access Review Operator Service Role	Lets you grant Access Review System app permissions to discover and revoke access as neede...	BuiltInRole	None	View
K2view Azure Application Installation	Provide the ability to install the K2view Azure Application	CustomRole	None	View

Showing 1 - 3 of 3 results.

Select the “Select members” link

Home > Subscriptions > K2view PAYG Marketing | Access control (IAM) >

Add role assignment

Role **Members** Conditions Review + assign

📘 Showing a filtered list of roles because your permissions include a condition. [Learn more](#)
[View my access](#)

Selected role K2view Azure Application Installation

Assign access to ☒ User, group, or service principal
☐ Managed identity

Members [+ Select members](#)

Name	Object ID	Type
No members selected		

Description Optional

If you have permissions, searching for the member will list the user on the list on the right. Select the user and press “Select”. When done, use the “Review and assign” button to apply the custom role assignment.

[Home](#) > [Subscriptions](#) > [K2view PAYG Marketing](#) | [Access control \(IAM\)](#) >

Add role assignment

[Role](#)
[Members](#)
[Conditions](#)
[Review + assign](#)

Showing a filtered list of roles because your permissions include a condition. [Learn more](#)
[View my access](#)

Selected role K2view Azure Application Installation

Assign access to ☒ User, group, or service principal
☐ Managed identity

Members [+ Select members](#)

Name	Object ID	Type
No members selected		

Description Optional

Review + assign

Previous

Next

Select members

Search by name or email address

No results.

Selected members:

No members selected. Search for and add one or more members you want to assign to the role for this resource.

[Learn more about RBAC](#)

Select

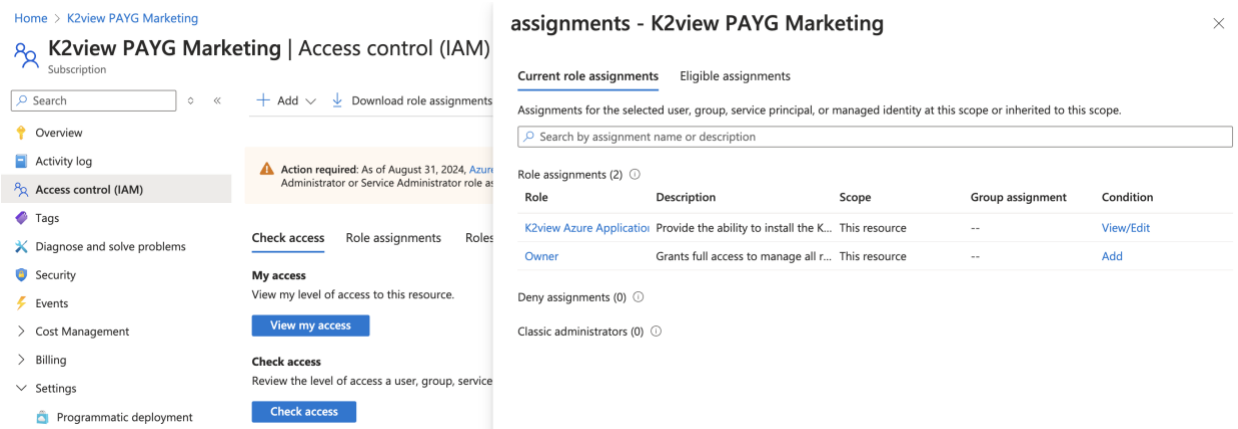
Close

These steps will have the affect of assigning the user with a custom role with these permissions:

"Microsoft.Authorization/roleAssignments/write",

"Microsoft.Authorization/roleAssignments/read",

"Microsoft.Authorization/roleAssignments/delete"



Home > K2view PAYG Marketing

K2view PAYG Marketing | Access control (IAM)

Subscription

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Security

Events

Cost Management

Billing

Settings

Programmatic deployment

Action required: As of August 31, 2024, Azure Administrator or Service Administrator role as

Check access

Role assignments

Roles

My access

View my level of access to this resource.

View my access

Check access

Review the level of access a user, group, service

Check access

assignments - K2view PAYG Marketing

Current role assignments

Eligible assignments

Assignments for the selected user, group, service principal, or managed identity at this scope or inherited to this scope.

Search by assignment name or description

Role assignments (2)

Role	Description	Scope	Group assignment	Condition
K2view Azure Application	Provide the ability to install the K...	This resource	--	View/Edit
Owner	Grants full access to manage all r...	This resource	--	Add

Deny assignments (0)

Classic administrators (0)

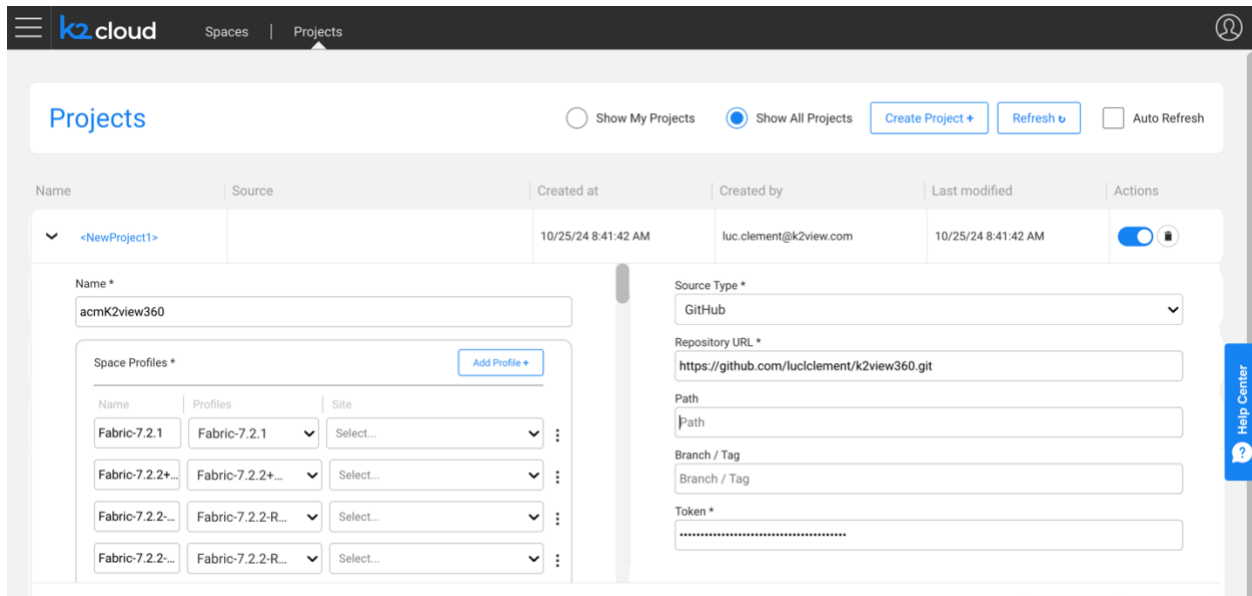
9.2. GitHub

Your organization may already have a Git repository. GitHub, GitLab and BitBucket are cloud providers that you can turn to should you like to get started.

During the course of this installation you will be asked to provide Git source code repository information including:

1. The provider: GitHub, GitLab, or BitBucket
2. A repository URL: in our example we used GitHub and this repository URL:
<https://github.com/luccllement/k2view360.git>
3. A branch: to keep things simple we used “main”
4. A Git Token: to provide the project access to the Git account

This information is captured on the project’s page



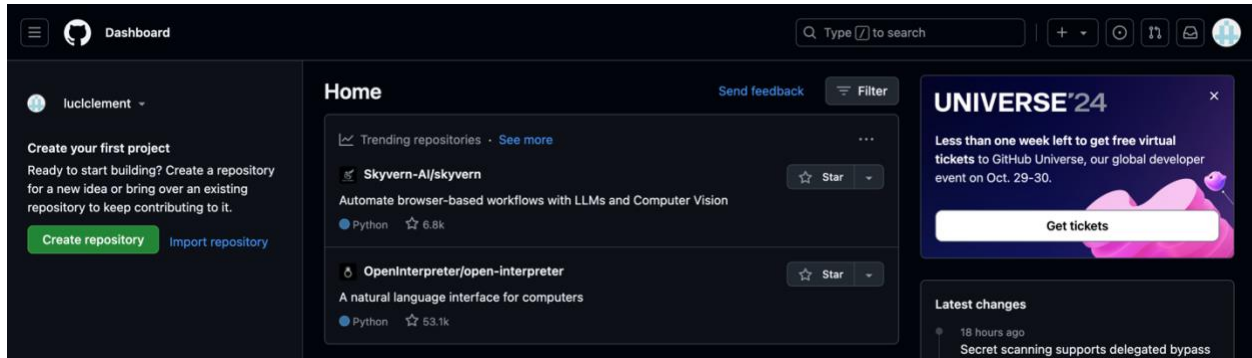
The screenshot shows the 'Projects' page in the K2view interface. At the top, there's a navigation bar with 'k2 cloud', 'Spaces', and 'Projects'. Below this, the 'Projects' section has tabs for 'Show My Projects' and 'Show All Projects', along with buttons for 'Create Project +', 'Refresh', and 'Auto Refresh'. A table lists projects, with the first one being 'acmK2view360'. Below the table, the configuration form for this project is visible. It includes fields for 'Name *' (acmK2view360), 'Source Type *' (GitHub), 'Repository URL *' (https://github.com/luccllement/k2view360.git), 'Path', 'Branch / Tag', and 'Token *'. There's also a 'Space Profiles *' section with a table of profiles and a 'Help Center' button on the right.

If you don't have one you can obtain from github.com. First you need to sign up at <https://github.com>

You will create a repository and then obtain a token.

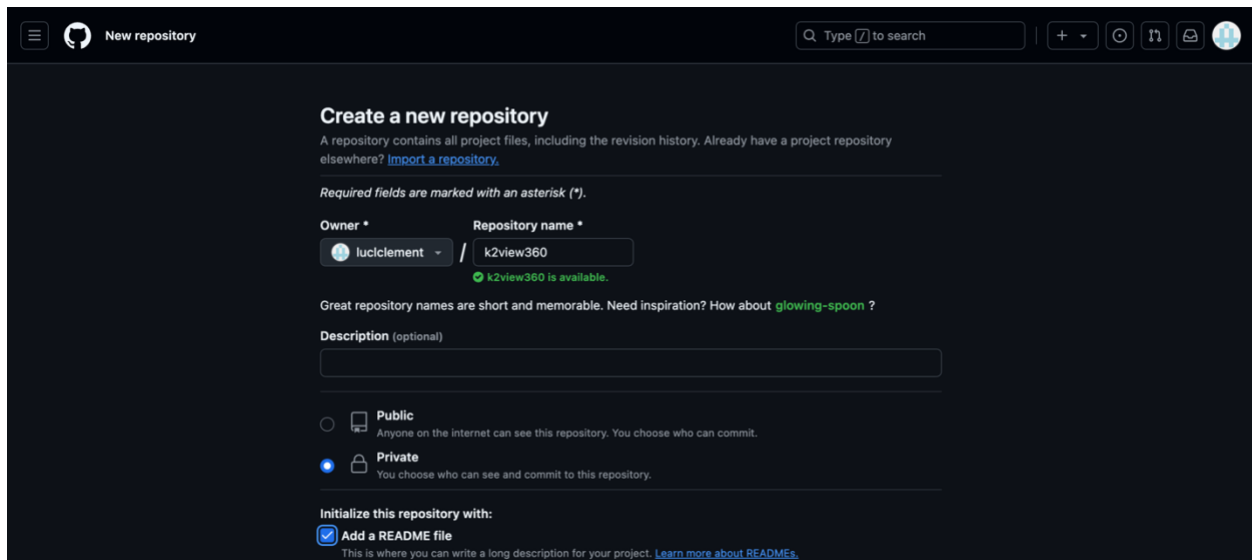
Create your repository

Click the “Create repository” button on the left side panel.

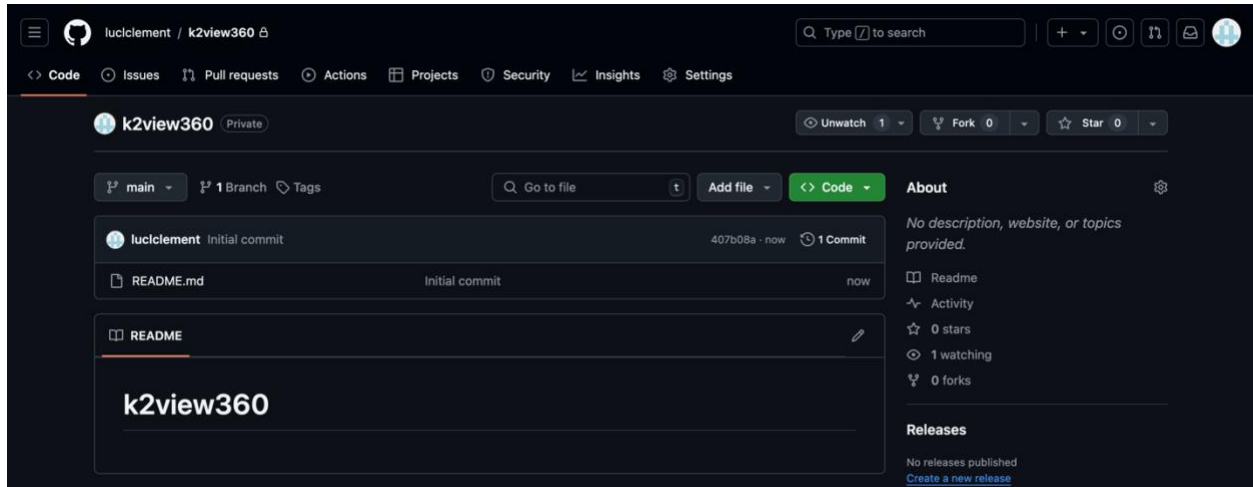


Give it a name. In this example we used k2view360.

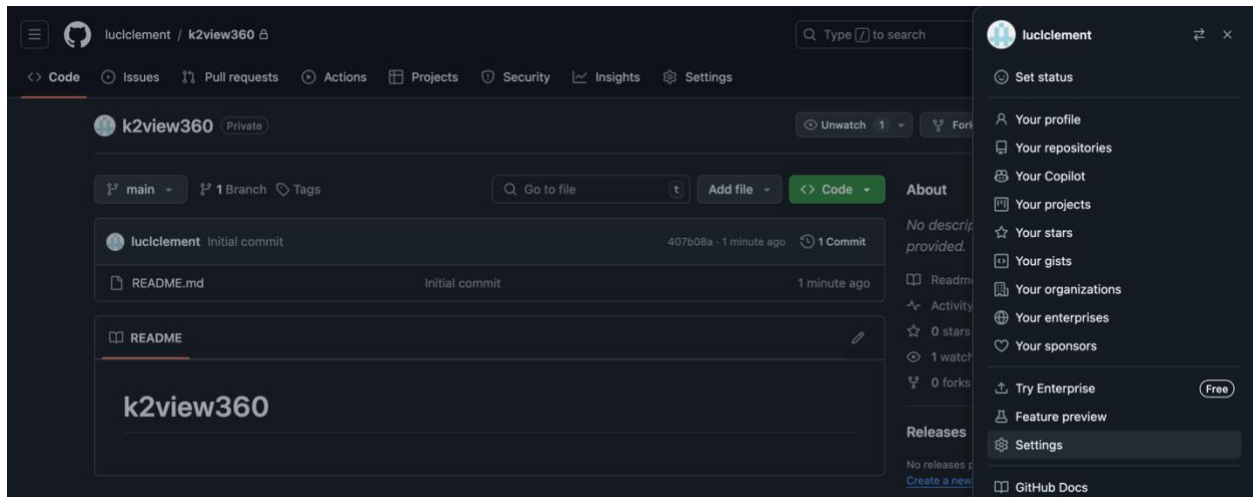
You should make it “Private”, and you should select the “Add a README file checkbox so that the repository is not empty.



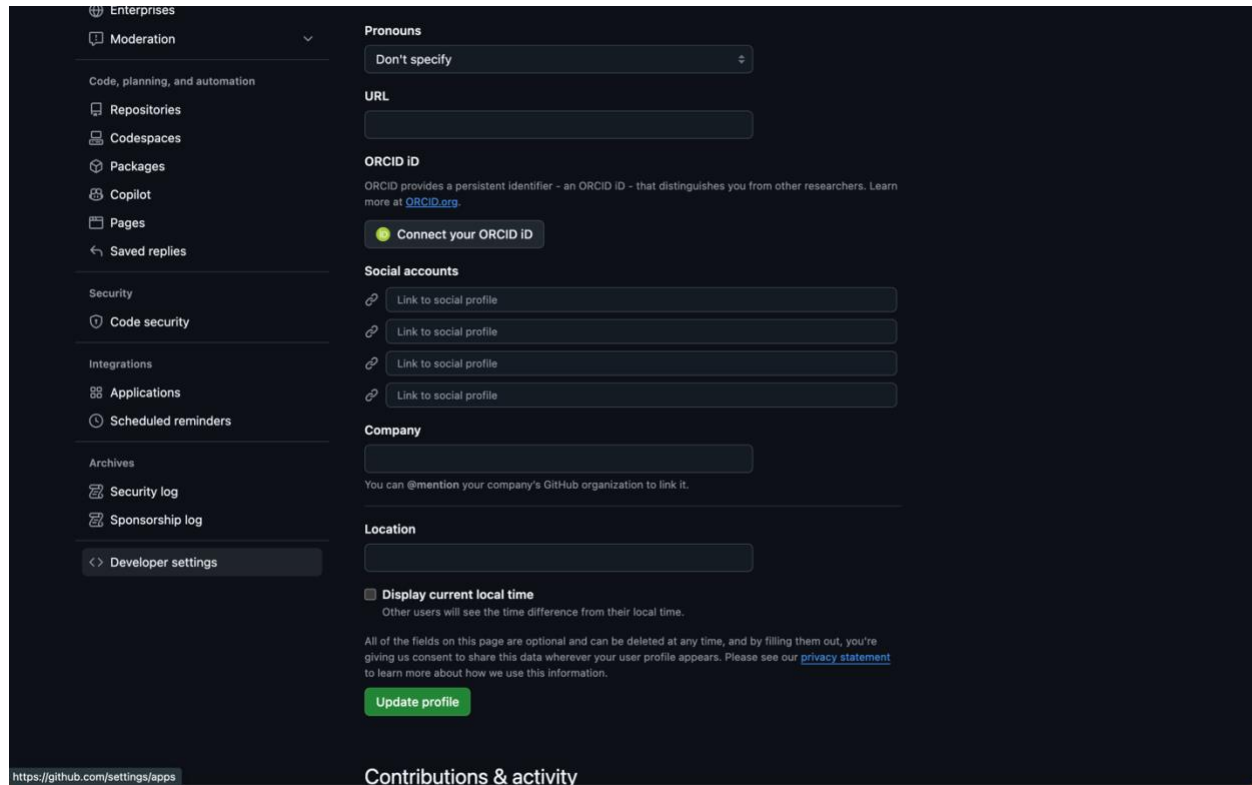
Once created you will find an empty README file.



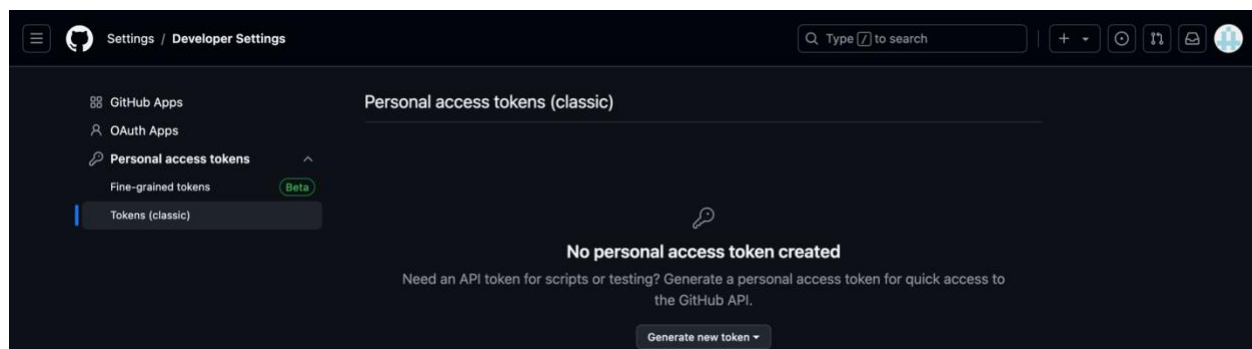
The next task is to obtain your Token. For this, drop down the user's icon on the top right of the page. Select Settings.



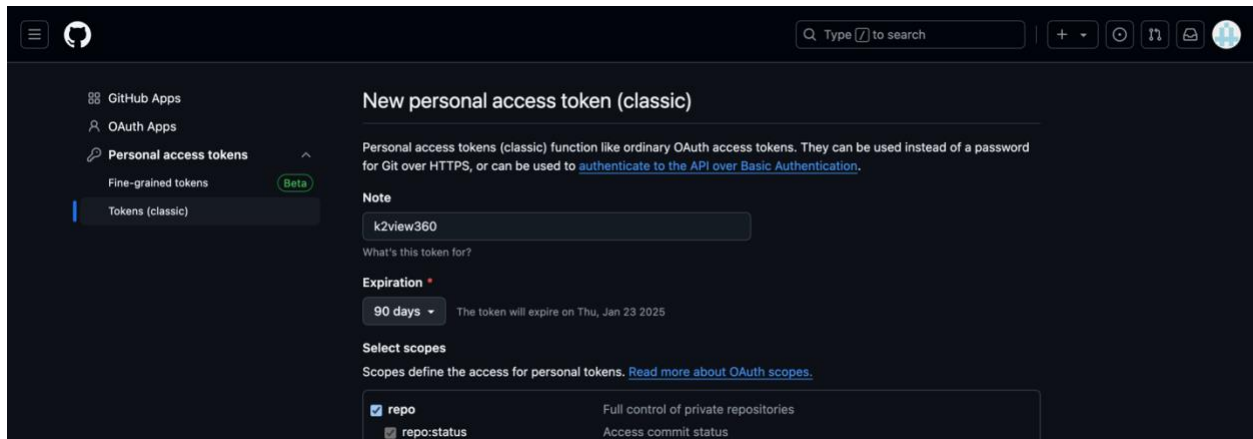
Then, from the left panel, select “Developer settings” on the bottom of the list.



Then, select “Personal access tokens” from the left panel, and select “Tokens (classic)” (if you’re adventurous, try the beta).



Select the “Generate new token” on the bottom of the page and create your token.



When the page refreshes you will be shown your token. Copy it to use in the Fabric Project creation page. Keep a copy of the token for safekeeping.

9.3. DNS zone

During the installation, a DNS zone will be created. This zone may take time to propagate into your environment and may not be available until it does.

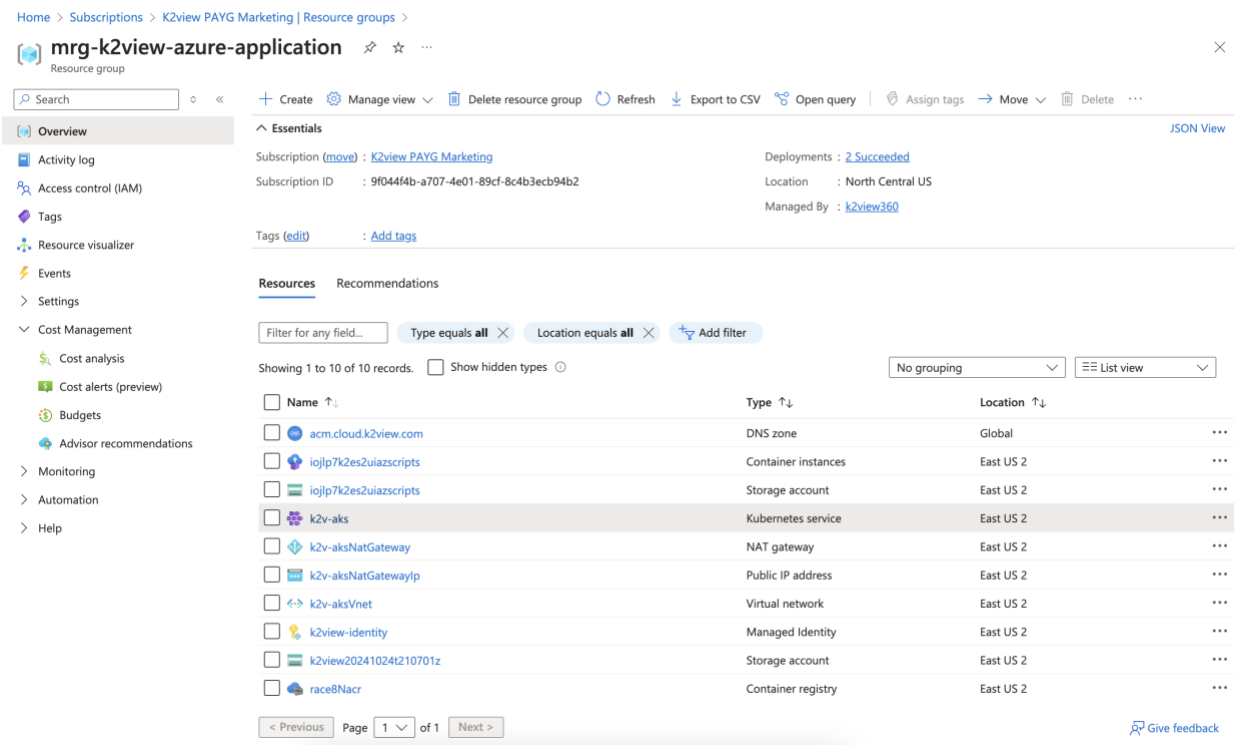
Some organizations may not allow the use of DNS record created in your installation's Public DNS zone to propagate outside the organization.

If that's the case, you will need to provide the address information to your organization to generate a DNS A record for the NGINX Ingress controller load balancer IP address created in our Public DNS Zone.

You should note that when creating a namespace, its name is associated as a subdomain to this domain name in the Ingress controller. For example, if the domain is "k2dev.company.com" and a created namespace is "test", then the URL of this namespace that users shall access will be "test.k2dev.company.com".

In our example, we will connect to a space accessible at acmk2view360-k2v-rnd.acm.cloud.k2view.com, a publicly resolvable domain.

To locate the address, select the Kubernetes service



Home > Subscriptions > K2view PAYG Marketing | Resource groups >

mrg-k2view-azure-application Resource group

Search

+ Create Manage view Delete resource group Refresh Export to CSV Open query Assign tags Move Delete

Overview Essentials JSON View

Activity log

Access control (IAM)

Tags

Resource visualizer

Events

Settings

Cost Management

Cost analysis

Cost alerts (preview)

Budgets

Advisor recommendations

Monitoring

Automation

Help

Subscription (move): [K2view PAYG Marketing](#) Deployments: [2 Succeeded](#)

Subscription ID: 9f044f4b-a707-4e01-89cf-8c4b3ecb94b2 Location: North Central US

Managed By: [k2view360](#)

Tags (edit): Add tags

Resources Recommendations

Filter for any field... Type equals all Location equals all Add filter

Showing 1 to 10 of 10 records. Show hidden types

No grouping List view

Name	Type	Location
acm.cloud.k2view.com	DNS zone	Global
iojlp7k2es2uiazsceipts	Container instances	East US 2
iojlp7k2es2uiazsceipts	Storage account	East US 2
k2v-aks	Kubernetes service	East US 2
k2v-aksNatGateway	NAT gateway	East US 2
k2v-aksNatGatewayIp	Public IP address	East US 2
k2v-aksVnet	Virtual network	East US 2
k2view-identity	Managed Identity	East US 2
k2view202410241210701z	Storage account	East US 2
race8Nacr	Container registry	East US 2

< Previous Page 1 of 1 Next >

Give feedback

Select the Services and ingresses. The address you are looking for is the ingress-nginx-controller's load balancer. In this example the address is 48.214.88.212.

Home > Subscriptions > K2view PAYG Marketing | Resource groups > mrg-k2view-azure-application > k2v-aks

k2v-aks | Services and ingresses ☆ ...

Kubernetes service

Search [] Create Delete Refresh Show labels Give feedback

Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Microsoft Defender for Cloud
Cost analysis
Kubernetes resources
Namespaces
Workloads
Services and ingresses ☆
Storage
Configuration
Custom resources
Events
Run command
Settings
Monitoring
Automation
Help

Services Ingresses

Filter by service name: Enter the full service name
Filter by namespace: All namespaces Add label filter

☐	Name	Namespace	Status	Type	Cluster IP	External IP	Ports	Age	
☐	kubernetes	default	Ok	ClusterIP	10.0.0.1		443/TCP	2 days	...
☐	kube-dns	kube-system	Ok	ClusterIP	10.0.0.10		53/UDP53/TCP	2 days	...
☐	metrics-server	kube-system	Ok	ClusterIP	10.0.66.7		443/TCP	2 days	...
☐	ingress-nginx-controller	ingress-nginx	Ok	LoadBalancer	10.0.149.85	48.214.88.212	80:31263/TCP4...	2 days	...
☐	ingress-nginx-controller...	ingress-nginx	Ok	ClusterIP	10.0.132.98		443/TCP	2 days	...
☐	ingress-test-service	ingress-nginx	Ok	ClusterIP	10.0.68.142		5678/TCP	2 days	...
☐	nginx-errors	ingress-nginx	Ok	ClusterIP	10.0.10.253		80/TCP	2 days	...
☐	fabric-service	acmk2view360-k2v-md	Ok	ClusterIP	10.0.55.51		3213/TCP	1 day	...

https://portal.azure.com/#@k2viewrmd.onmicrosoft.com/resource/subscriptions/9f04df4b-a707-4e01-88cf-Bc-d53ech9d52/resourceGroups/mrg-k2view-azure-application/providers/Microsoft.ContainerService/managedClusters/k2v-aks/serv...

9.4. Certificates

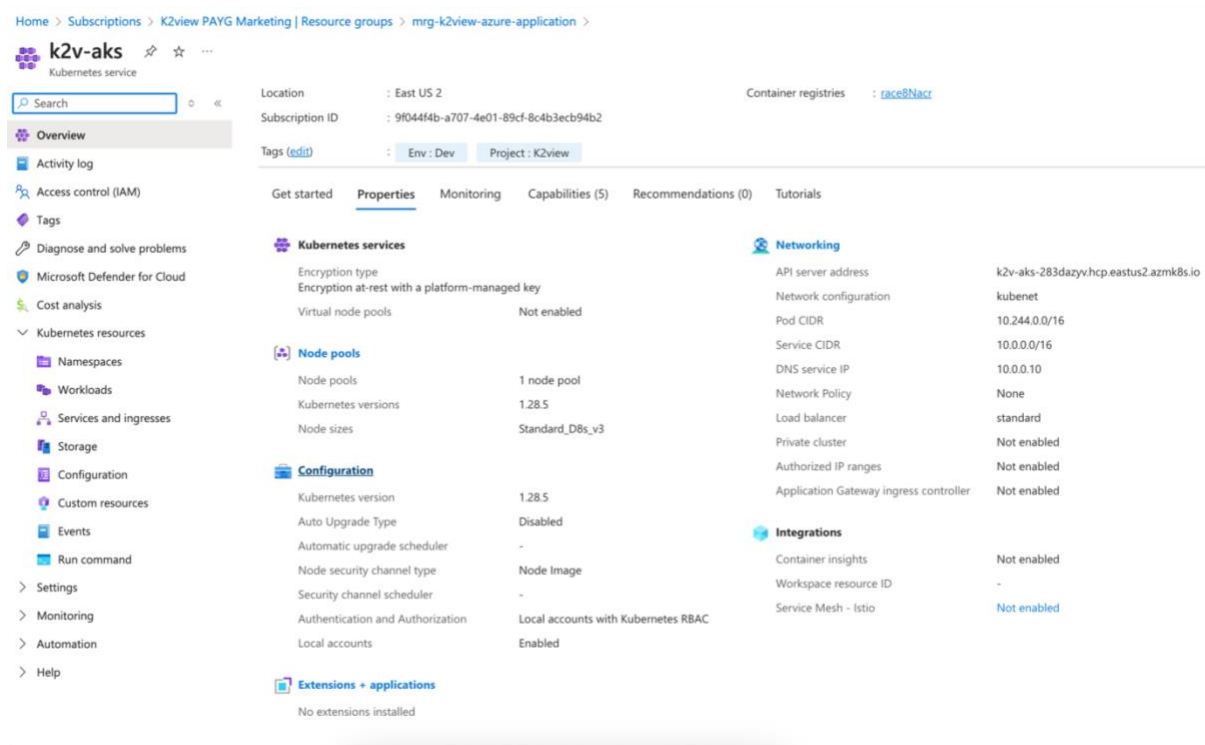
Expiration

NOTE: Certificates have expiration dates. Please note the date of expiration of the certificate provided to you and replace it before the expiration date as described next.

Replacing Certificates

If you encounter certificate issues or need to update your certificate, follow these instructions. You can also make a support request with K2view for assistance.

First, select the Managed Azure Application resource group created by the installation. Select the Kubernetes service item.



The screenshot shows the Azure portal interface for a Kubernetes service named 'k2v-aks'. The breadcrumb navigation path is: Home > Subscriptions > K2view PAYG Marketing | Resource groups > mrg-k2view-azure-application > k2v-aks. The left sidebar contains a search bar and a list of navigation items: Overview (selected), Activity log, Access control (IAM), Tags, Diagnose and solve problems, Microsoft Defender for Cloud, Cost analysis, Kubernetes resources (expanded), Namespaces, Workloads, Services and ingresses, Storage, Configuration, Custom resources, Events, Run command, Settings, Monitoring, Automation, and Help. The main content area displays the 'Properties' tab for the 'k2v-aks' service. At the top, it shows the Location as 'East US 2', Subscription ID as '9f044f4b-a707-4e01-89cf-8c4b3ecb94b2', and Container registries as 'raced@Nacr'. Below this, there are tags for 'Env: Dev' and 'Project: K2view'. The 'Properties' tab is divided into several sections: 'Kubernetes services' (Encryption type: Encryption at-rest with a platform-managed key, Virtual node pools: Not enabled), 'Node pools' (Node pools: 1 node pool, Kubernetes versions: 1.28.5, Node sizes: Standard_D8s_v3), 'Configuration' (Kubernetes version: 1.28.5, Auto Upgrade Type: Disabled, Automatic upgrade scheduler: -, Node security channel type: Node Image, Security channel scheduler: -, Authentication and Authorization: Local accounts with Kubernetes RBAC, Local accounts: Enabled), 'Networking' (API server address: k2v-aks-283dazyvhcp.eastus2.azmk8s.io, Network configuration: kubenet, Pod CIDR: 10.244.0.0/16, Service CIDR: 10.0.0.0/16, DNS service IP: 10.0.0.10, Network Policy: None, Load balancer: standard, Private cluster: Not enabled, Authorized IP ranges: Not enabled, Application Gateway ingress controller: Not enabled), 'Integrations' (Container insights: Not enabled, Workspace resource ID: -, Service Mesh - Istio: Not enabled), and 'Extensions + applications' (No extensions installed).

Then select the “Configuration” left menu item as shown here, select the Secrets tab, and then select the “wildcard-certificate”.

Home > Subscriptions > K2view PAYG Marketing | Resource groups > mrg-k2view-azure-application > k2v-aks

k2v-aks | Configuration ☆ ...

Kubernetes service

Search [] Create Delete Refresh Show labels Give feedback

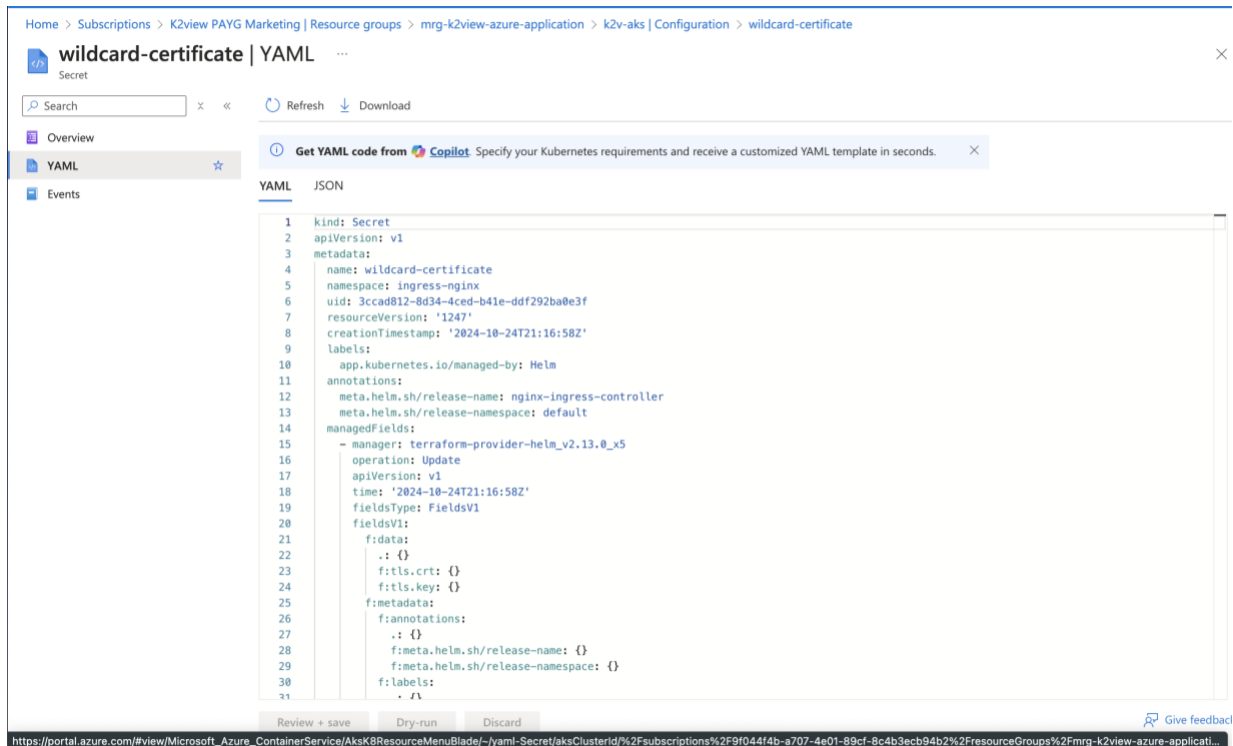
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Microsoft Defender for Cloud
Cost analysis
Kubernetes resources
Namespaces
Workloads
Services and ingresses
Storage
Configuration
Custom resources
Events
Run command
Settings
Node pools
Cluster configuration
Application scaling
Networking
Extensions + applications

Config maps **Secrets**

Filter by secret name: Filter by namespace: Add label filter

☐	Name	Namespace	Type	Data	Age
☐	connectivity-certs	kube-system	Opaque	3	22 hours
☐	bootstrap-token-76dr0f	kube-system	bootstrap.kubernetes.io...	4	22 hours
☐	sh.helm.release.v1.k2view-agent.v1	default	helm.sh/release.v1	1	22 hours
☐	agent-config-secrets	k2view-agent	Opaque	5	22 hours
☐	k2view-agent-secret	k2view-agent	kubernetes.io/service-ac...	3	22 hours
☐	sh.helm.release.v1.nginx-ingress-controller.v1	default	helm.sh/release.v1	1	22 hours
☐	wildcard-certificate	ingress-nginx	kubernetes.io/tls	2	22 hours
☐	ingress-nginx-admission	ingress-nginx	Opaque	3	22 hours
☐	common-env-secrets	acmk2view360-k2v-rnd	Opaque	12	3 hours
☐	config-init-secrets	acmk2view360-k2v-rnd	Opaque	6	3 hours
☐	config-secrets	acmk2view360-k2v-rnd	Opaque	5	3 hours
☐	secret-deploy-apikey	acmk2view360-k2v-rnd	Opaque	1	3 hours

After opening the wildcard certificate and selecting the YAML section, scroll down and locate the Key and Certificate sections and paste replacement values (not shown here). These values are the base64 encoded values of your PEM certificate and private key.



Your PEM TLS certificate and its associated private key

Your network team must create a wildcard TLS certificate for the domain you selected as your Domain Ingress. The full certificate authority chain needs to be present. During the installation, you must paste the base64-encoded values of your TLS certificate and its associated private key.

- **TLS Private:** the base64-encoded private key associated with the TLS certificate
- **TLS Certificate:** a base64-encoded wildcard certificate. For example, for the domain ingress of acm.cloud.k2view.com you need to create a wildcard certificate for *.acm.cloud.k2view.com for hosts of this domain.

What you paste in should look like the following:

base64-encoded TLS Private Key:

```
LS0tLS1CRUdJTiBQUk1wQVRFIEtFWS0tLS0tCk1JSUpRd01CQURBTk1na3Foa21HOXcwQkFRRUZBQVNDQ1Mwd2dna3BBZ  
OVBQW9JJQ0FRRE9yakUrNONQVku2QVgKV01FRFVvdnVLC01XOEJ4MTR  
L...  
QT1ZV1pKKz1qRkhFUTdBZXB3S1RtMEVraz0KLS0tLS1FTkQgUjJkFURSBLRVktLS0tLQo=
```

base64-encoded TLS Certificate:

```
LS0tLS1CRUdJTiBDRVJUSUZJQ0FURSB0tLS0tCk1JSUdGRENDQ1B5Z0F3SUJBZ01TQS9YVFIKaUpXOXMzb1dZZ1N5a0RpZ  
1h5TUeWR0NTcUdTSWlZRFFFQkN3VUEKTURNeEN6QUpCZ05WQkFZVEFsV1RNU113RkFZRFZRUUtFdzFNW1hRbm  
...  
1ND0KLS0tLS1FTkQgQ0VSVE1GSUNBVEUtLS0tLQ==
```

Note: The PEM certificate and key must be further encoded using base64 as input to the installation screen. To encode a PEM certificate (e.g., cert.pem) to base64 on both Linux and Windows, you can use the following commands.

Linux: You can use the base64 command to read the cert.pem file, encodes it in base64, and saves the output to cert_base64.txt.

```
base64 cert.pem > cert_base64.txt
```

Windows: On Windows, you can use the certutil command to encode the cert.pem file in base64 and saves the result to cert_base64.txt.

```
certutil -encode cert.pem cert_base64.txt
```

Prerequisite: You should prepare these base64 encoded files before the start of the installation.

10. Troubleshooting

DNS Issues

To start, please refer to the [DNS zone](#) section. For assistance, please make a support request with K2view.